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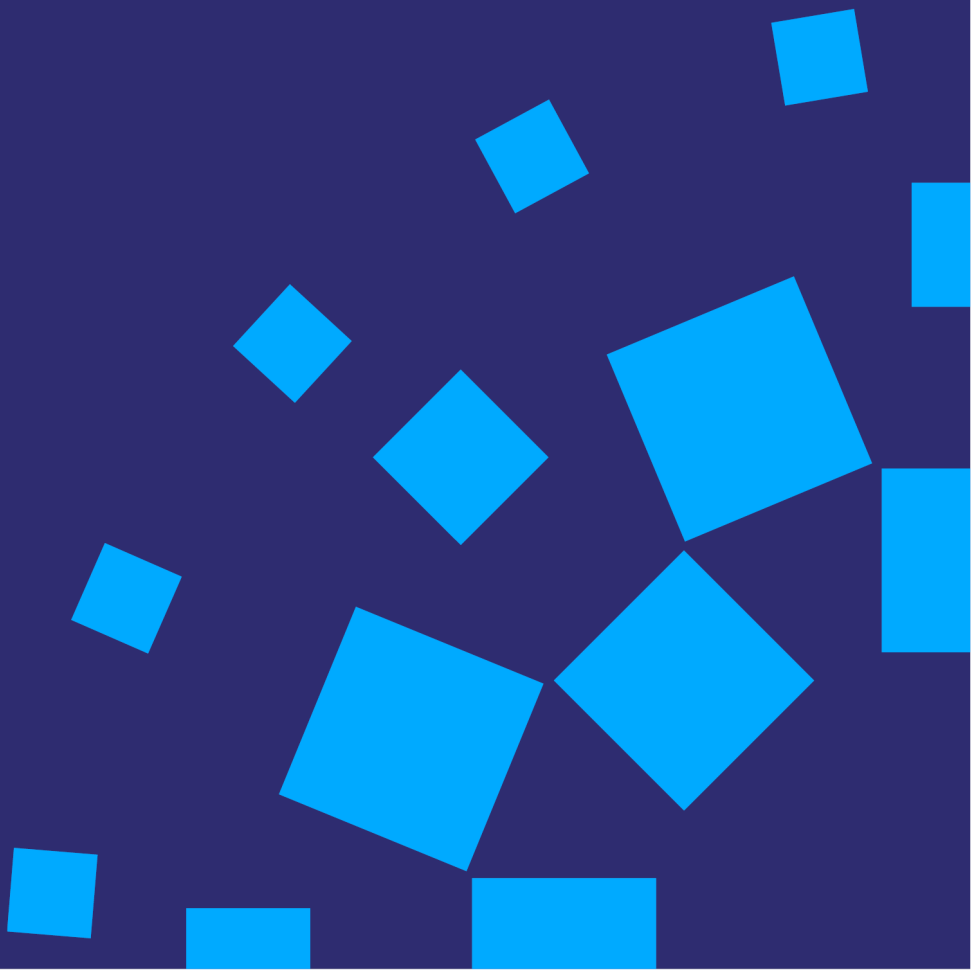
Evaluation of a forensic marking intervention for domestic abuse

Technical report

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Executive summary

Key findings

The forensic marking intervention aimed to reduce harm from domestic abuse by:

- increasing feelings of safety and freedom for victims
- increasing trust and confidence in the police
- deterring repeat offending
- bringing perpetrators to justice through supporting the detection of crimes

To understand the impact of the intervention, the evaluation compared repeat offending for those who received the forensic marking products to a matched comparison group. We also conducted a longitudinal survey of victims, interviews with victims and police officers, and case studies of activations.

- The forensic marking intervention was associated with a reduction in repeat incidents over a six-month follow-up period for the intervention group, compared to a matched comparison group. While this was not significant, there was encouraging preliminary evidence, with a reduction factor of around 0.78. This equates to a reduction of 22% in the number of crimes that would have happened without the intervention in place.
- The cost of delivering the intervention is estimated to be around £131.32 per person. Depending on the method used, the benefit–cost ratio was estimated to be between 38.7 and 82.8, meaning that for every pound spent, there would be an economic benefit of between £38.70 and £82.80.
- Survey and interview data showed a consistent increase in feelings of safety and freedom for victims who had received the intervention, which was maintained over time. Most victims said this was to some extent, or to a great extent, due to the forensic marking device. Many victims felt confident to resume activities outside of their homes that they had stopped doing.
- In the self-report survey, 22 of the 49 victims (45%) had no further unwanted contact at all in the six-month follow-up period. For those that did, unwanted contact was frequently through another person. Most victims with unwanted contact reported this to the police for some or all of the incidents.

- Interviews with victims showed that experiences varied in relation to further unwanted contact and whether forensic marking (and the warning given to the perpetrator) was linked to this. Some noted a cautious optimism that perpetrator could no longer ‘get away’ with further contact and bail or order breaches. However, a small number of victims remained unconvinced, given the failure of previous interventions.
- The victim survey and interview data showed an initial increase in terms of trusting the police to respond appropriately, but this was not maintained over time. Satisfaction with response and confidence to report future incidents also reduced over time.
- There were only five activations during the six-month follow-up period. While the forensic marking evidence did not form a key part in any of the prosecutions, it did form part of the investigation process and was thought to have reduced investigation time and strengthened cases.
- A warning was only recorded as being given to 60% of perpetrators, so the deterrence effect may be greater if this is delivered more consistently. In four of the five activations, the warning had not been given to the perpetrator.

Background

There is a need to understand the potential of situational crime prevention methods to reduce domestic abuse, as underlined by the:

- significant harms caused by domestic abuse
- high demand it places on the criminal justice system
- high rate of attrition through the system

Existing research is limited but suggests potential. The College of Policing secured funding from the Cabinet Office Evaluation Accelerator Fund to evaluate police interventions focused on reducing violence against women and girls (VAWG). One of these was forensic marking. Forensic marking is a situational crime prevention method that marks a victim’s property using a chemically marked solution. This solution is unique to each victim, invisible until under ultraviolet (UV) light, and either grease-based or water-based. It can also be deployed via a hand-held canister to link a domestic abuse perpetrator to the scene of an incident.

A theory of change was developed collaboratively through online facilitated group discussions and a review of existing literature. The identified outcomes were:

- increased trust and confidence in policing
- increased feelings of safety and freedom for victims
- a reduction in domestic violence and abuse
- an increase in repeat perpetrators brought to justice

Research methods

The evaluation aimed to:

- assess to what extent the intervention met the theory of change outcomes
- assess whether the intervention was cost-effective
- understand how the intervention was implemented in each of the five police forces participating in the evaluation

The research comprised:

- a mixed methods, longitudinal study using police incident data (with a comparison group matched by fitting a propensity score using a logistic regression model)
- a victim self-report survey
- in-depth interviews with:
    - 36 police officers
    - 15 non-police stakeholders (project partners, specialist domestic abuse organisations)
    - 27 victims who were given a forensic marking device (some victims were interviewed more than once, with 35 interviews in total)

Ethical approval was granted for all methods by the Durham University Department of Sociology Research Ethics Committee.

Implementation findings

Each force had a slightly different process for managing the forensic marking intervention, depending on the local setup and responsibility for giving out the intervention.

All forces provided criteria or guidelines to help officers decide who should receive the forensic marking products, although these guidelines differed slightly between forces. Nearly all of those offered a forensic marking intervention were assessed as high or medium risk. Stalking was the most frequent crime type that led to the intervention being offered (in more than four in ten cases). The spray canisters were offered more often than the grease. The spray canisters were cheaper, were not affected by the weather and were portable, so they could be taken out of the house. However, some police officers and victims felt that it looked like a pepper spray. Both officers and victims preferred the smaller canisters to the larger ones.

In the interviews with victims, almost all described the process by which they were given the device in positive terms. Overall, 48 out of 49 victims responding to the survey said that they agreed or strongly agreed with the statement “I would be confident using the spray”.

A key part of the intervention is to deter the perpetrator from further offending. The perpetrator is warned that the victim has received forensic marking products, which could be used to provide proof that the perpetrator has been in close contact. However, this warning was only recorded as being given in around 60% of cases. As part of the intended deterrence effect, victims were offered stickers or posters to put on their doors and windows warning that forensic products were in use. In practice, a large proportion felt uncomfortable about making their victimisation more visible in their community.

Outcomes

Victim data showed a consistent increase feelings of safety and freedom, which was maintained over time. There was also an initial increase in trust in the police to respond appropriately, but this was not maintained over time.

Police data was analysed on further offences within the same victim–perpetrator pair. This showed an association between the forensic marking intervention group and a reduction by a factor of approximately 0.78 in incidents over the following six months, compared to the matched comparison group. This was not statistically significant, but was encouraging preliminary evidence of a reduction.

In the self-report survey, 22 of the 49 victims (45%) did not have any further unwanted contact at all in the six-month follow-up period. For those that did, the

most frequent form of unwanted contact was through another person. Most victims with unwanted contact continued to report this to the police for some or all of the incidents.

During the course of the evaluation period, there were very few activations of the forensic marking products. While the forensic marking evidence did not form a key part in any of the prosecutions that followed the activation incidents, it did contribute to the investigation process by reducing investigation time and strengthening cases.

Limitations

There are a number of evaluation limitations, including:

- a lack of consistency in how the intervention was delivered
- the inconsistent use and recording of the perpetrator warning
- problems with police data
- a high rate of attrition in the victim self-report survey

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1. Introduction

1.1. Background

Domestic abuse has significant, often long-term, impacts on its victims and makes up a high proportion of police workload. Despite this, a minority of offenders are brought to justice (through the criminal justice system or otherwise). In the year ending March 2024, over 1.3 million (1,350,428) domestic abuse-related incidents were recorded by the police in England and Wales. Of these, 850,000 were recorded as crimes (Office for National Statistics (ONS), 2024a).

However, only a very small proportion of offences resulted in positive charging decisions and subsequent convictions. In total, 51,183 prosecutions were recorded in the year ending March 2024, with three-quarters of prosecutions resulting in a conviction (McPhee and others, 2022; ONS, 2024b).

There is a need to understand the potential of situational crime prevention methods to reduce domestic abuse, as underlined by the:

- significant harms caused by domestic abuse
- high demand it places on the criminal justice system
- high rate of attrition through the system

Despite evidence supporting the effectiveness of situational crime prevention methods for various crime types, their application in the context of domestic abuse remains under-developed. Prenzler and Fardell (2017) conducted an international literature review of security-based programmes aimed at preventing domestic abuse. They found evidence of a reduction in violence for five types of interventions:

- offender GPS tracking
- shelter security
- home security
- personal duress alarms or panic alarms
- home security and panic alarms combined

This review included the Hester and Westmarland (2005) analysis of the ‘Bradford Staying Put’ project in England, funded through the Home Office crime reduction

programme. This project found positive outcomes in relation to the distribution of mobile phones, home security upgrades and panic alarms.

Building on these outcomes, the College of Policing sought to develop the limited evidence base on what works to prevent violence against women and girls (VAWG).

1.2. Intervention

1.2.1. Forensic marking

Forensic marking is a situational crime prevention method. It uses a unique, chemically marked grease-based or water-based solution, which is invisible to the human eye until placed under ultraviolet (UV) light. It has traditionally been used in relation to property theft, with owners marking an item of property, such as a bike, to:

- deter its theft
- identify who stole it
- help reunite the stolen property with its owner

This project extended the use of forensic marking to a domestic abuse context, focusing on the following:

- **Detection**: Forensic marking could be used by a victim to link a suspect to the location of an incident. The victim could mark the suspect who was not permitted to approach them by using a hand-held canister spray or applying transferable grease to their property.
- **Deterrence**: The perpetrator is warned that the victim has been provided with forensic marking products, and that an activation of the product could provide evidence linking them to a location or the victim. This is intended to deter the perpetrator from approaching the victim and their property.

1.2.2. Implementation

Five police forces initially agreed to take part in the trial. Four of the forces were already using forensic marking for domestic abuse, and one was looking to start.

Products used by forces were provided by two companies, Selecta DNA and Detertech, which provided products under the brand name SmartWater.

There were two main types of forensic marking products available to forces. :

- **Forensic spray**: A canister that can deploy a spray of uniquely coded liquid to mark an offender several metres away.
- **Forensic grease or gel**: A substance used to mark objects such as door handles, vehicles or garden furniture and that will transfer to an offender on touch.

The forces decided which product was appropriate for each victim. The most common product was the spray canister that the victim could carry with them. There were also a small number of fixed forensic marking units that could be installed in a victim's home next to the door. These units could be activated by a remote control if the perpetrator entered the property. The units were not portable and were seldom used. Victims who were issued with forensic marking products were usually offered warning stickers or a home protection kit. This kit included a UV pen for property marking and a warning sticker or poster to display in their window or on their door.

In addition to the forensic marking products, forces expected police officers or staff to give in-person warnings to perpetrators or, if not possible, send a warning letter to them. The warning explains that the victim had received forensic marking products that will allow the police, through forensic evidence, to place the perpetrator at a specific location.

1.2.3. Deploying forensic marking products

Each force had a slightly different process to decide who should receive the forensic marking kits in practice. Some forces had more formal criteria, while others provided them on a more ad hoc basis. In all forces, the intervention was only considered suitable for use in cases where the perpetrator was no longer in a relationship with, or cohabiting with, the victim. This is because forensic marking is intended to deter perpetrators from approaching the property or the victim, and to provide evidence if they did. The evaluation only included cases of intimate partner violence (rather than wider forms of domestic abuse). However, some forces used the forensic marking for other forms.

A range of criteria were considered when selecting appropriate cases to use the forensic marking intervention. Decisions were based on the professional judgement of the police officer attending and on the victim's circumstances, rather than more formal measures. The criteria included:

- risk assessment, likely to be medium or high due to other criteria around reoffending
- likelihood of reoffending
- level of harm if reoffending occurs
- victim engagement and agreement
- environmental and other factors (for example, drug or alcohol use that may affect use of the intervention, suitability of the property for displaying signs or using the grease)
- victim address known to the perpetrator

1.2.4. Costs

The intervention was relatively low-cost and quick to implement. Costs for the forensic marking spray canisters were around £50 per item, which could be returned and redeployed if not used. Costs for the grease ranged from £110 to £140 per pot, which had a single use with one property. Other material costs included stickers or posters, plus the cost of printing information packs and paperwork that needed to be completed.

1.3. Evaluation

The evaluation assessed the extent of the impact of the forensic marking intervention on the harms associated with domestic abuse, drawing on a number of outcome measures. It also looked at whether forensic marking is a cost-effective approach to use. A process evaluation was conducted to understand how the intervention was implemented in each force in practice.

1.3.1. Theory of change

The theory of change describes how each element of an intervention is expected to lead to the intended outcomes and overall impact. The theory of change for this intervention (Table 1) was developed collaboratively in a workshop with:

- police officers
- key stakeholders from the Home Office and the College of Policing
- members of the research team

The theory of change was also informed by a review of existing literature. This included collaborative research between academics and West Yorkshire Police and South Yorkshire Police, where the initiative began. This research found that domestic abuse victims who were given SmartWater (a brand of forensic marking) felt that it had a positive impact on their wellbeing (Whitfield and others, 2022). Specifically, victims reported feeling safer, feeling reassured and having greater confidence in the police (Whitfield and others, 2022).

Table 1: Theory of change for forensic marking.

Interventions	▪ Forensic marking alongside perpetrator warning.
Outputs	▪ Number of kits deployed. ▪ Number of victims engaged. ▪ Number of perpetrators visited. ▪ Number of victims reporting improvements in trust and confidence in policing. ▪ Number of victims reporting improvements in feelings of safety. ▪ Number of repeat incidents. ▪ Number of charges and convictions following repeat incidents.
Outcomes	▪ Increased trust and confidence in policing. ▪ Increased feelings of safety and freedom for victims. ▪ Reduction in domestic violence and abuse. ▪ Increase in repeat perpetrators brought to justice.
Impacts	▪ Reduction in the harms associated with domestic abuse.

The intervention – the forensic marking alongside perpetrator warning – was intended to have an impact by reducing the harms associated with domestic abuse. Previous research and practitioners who had used forensic marking before suggested that victims receiving the intervention would feel safer and freer, and would also have greater trust and confidence in the police. Participants in the theory of change workshops thought this would work in two ways:

- Through the deterrent effect of the warning, perpetrators would be less likely to approach the victim, knowing something was in place. The expected outcome from this would be a decrease in reoffending by the perpetrator.
- The intervention would work by reducing the investigation time or increasing guilty pleas of those who were marked by the spray or the grease. The outcome from this is an increase in the number of perpetrators being brought to justice.

The theory of change suggested that these two factors would make victims feel that the police were doing everything that they could to help them. This would lead to an increase in trust and confidence in the police, which can be measured as another outcome. The victims would then feel more able to leave the house knowing that, in the case of the spray, they had something that they could carry with them. The expected outcome from this would be an increase in the victims' feelings of safety and freedom.

1.3.2. Evaluation aims

The evaluation aimed to understand the extent to which the outcomes from the theory of change were achieved through the intervention. As listed in Table 1, the intended outcomes were:

- increased feelings of safety and freedom for victims
- increased trust and confidence in policing
- reduction in domestic abuse
- increase in repeat perpetrators being brought to justice

1.3.3. Evaluation approach

A quasi-experimental mixed-method longitudinal design was used to evaluate the impact of the intervention. The quasi-experimental approach was used to compare:

- reoffending rates by perpetrators associated with victims who received forensic marking (the intervention group)
- a matched group identified from domestic abuse cases that occurred when forensic marking was not available (the comparison group)

Propensity score matching was used to identify cases to include in the comparison group that would have been likely to receive forensic marking if it had been available.

A longitudinal survey of the intervention group victims was used to understand outcomes of trust and confidence in the police and feelings of safety and freedom over time.

Other evaluation designs were considered. The decision was made not to use a randomised controlled trial. This is because there was reluctance from police force leads to withhold the intervention from victims who could benefit from it over the whole duration of the evaluation. Pausing the intervention for two weeks to collect comparison group data coincided with a relaunch of the intervention in one force and a change of supplier in another. The third force did not start the intervention until later in the evaluation period, giving a longer time period when the intervention was not yet available to identify comparison cases. Practicalities in managing the randomisation process would have been difficult, as local processes differed in terms of decision making, approvals and distribution.

2. Research methods

2.1. Data collection and analysis

The research consisted of a quasi-experimental mixed-methods, longitudinal study. The study used:

- police incident data
- a victim self-report survey
- in-depth interviews with:
  - police officers
  - non-police stakeholders (project partners, specialist domestic abuse organisations)
  - victims who were given a forensic marking device

Views on forensic marking were also included from domestic abuse perpetrators, but these were not the perpetrators linked to the victims who were given the forensic marking devices. Table 2 summarises the approaches used for the evaluation.

Ethical approval was granted for all methods by the Durham University Department of Sociology Research Ethics Committee.

Table 2: Summary of research methods used for each outcome .

Research method	**Longitudinal survey of victims**	**Interviews**	**Analysis of police-recorded crime data**	**Case studies of activations**
Sample size	Five surveys over six months (survey 1, n = 96; survey 5, n = 49)	Victims (n = 35), police (n = 37), stakeholders (n = 15)	Three forces: intervention group (n = 232), comparison group (n = 276)	n = 5
Outcomes				
Feelings of safety and freedom for victims	Yes	Yes	No	Yes
Trust and confidence in policing	Yes	Yes	No	Yes
Reduction in domestic violence and abuse	Yes	Yes	Yes	Yes
Repeat perpetrators brought to justice	No	No	Yes	Yes

Research method	**Longitudinal survey of victims**	**Interviews**	**Analysis of police-recorded crime data**	**Case studies of activations** (n = 5)
	Five surveys over six months (survey 1, n = 96; survey 5, n = 49)	Victims (n = 35), police (n = 37), stakeholders (n = 15)	Three forces: intervention group (n = 232), comparison group (n = 276)	
Process				
Project implementation	No	Yes	No	No
Cost-effectiveness	No	Yes	Yes	No

2.1.1. Police crime and incident data

A quasi-experimental design was used to compare reoffending outcomes between the intervention group and a matched comparison group. This was to see if there had been a reduction in domestic abuse offences between the same victim–perpetrator pair during the six-month period following the reference incident

For the intervention group, police forces provided data about:

- which victims received forensic marking
- which products victims received
- details of the offence leading to the intervention being provided
- details of crimes and incidents committed by the perpetrator over the previous 12 months
- whether protection orders or bail conditions were put in place following the incident, to understand other potential deterrence factors that might influence further offending by the perpetrator

The analysis of police data focused on cases with a female victim, and on cases where the victim was assessed as being high- and medium-risk. Further details of the characteristics of the victims receiving forensic marking are provided in the **Managing and administering the intervention** section and **Appendix 11**.

Identifying the comparison group

To identify the comparison group, forces provided similar data for all intimate partner domestic abuse cases that were recorded during a two-week period when the intervention was not running. For forces already implementing the forensic marking intervention, this was paused for two weeks. For the force that had not yet started the intervention, the comparison data was collected from the same two-week period as other forces, before the intervention started.

Approach to propensity score matching

Propensity score matching is a quasi-experimental approach to evaluation. Using police-recorded crime data, a comparison group is created consisting of victims who did not receive the intervention. These victims in the comparison group are closely matched to the intervention group across a number of characteristics. As a result,

they are considered to be equivalent to the intervention group before the intervention took place. Differences seen between the groups after the intervention has been implemented are likely to be due to the effect of the intervention.

The comparison and intervention data were combined and used to calculate a propensity score for each victim. This score is the estimated probability of a particular victim being given some forensic marking equipment, based on their characteristics. Propensity score models were fitted separately for each police force using a logistic regression model (see **Appendix 12** for more detail). Identifying matched cases from a different force area would not be appropriate, given the variability in managing domestic abuse cases between forces. This includes variability in assessing and managing risk, and in the support services available to victims.

For all forces, there were a large number of cases that had very low propensity scores, which were removed from the data set. The full comparison group for each force contained all cases for which the propensity score was within the range of the intervention group propensity scores. In total for the impact analysis, there were 232 intervention cases and 276 comparison cases (see **Appendix 13**).

To create balanced groups for the analysis, the comparison and intervention groups were randomly sampled from to ensure equal numbers of cases within each 0.2 interval of propensity scores. This was repeated 50 times, creating 50 balanced subsamples, to account for sensitivity to the sampling. Each balanced data set had 378 cases.

Impact analysis – reoffending rates

For both the intervention group and the matched comparison group, police data was used to identify individual offenders who committed index offences against individual victims. The data was used to track whether the perpetrators committed further offences against the same victims over a six-month period following implementation of forensic marking. The analysis assessed whether there was a difference in the number of further offences with the same victim–perpetrator pair between the comparison and intervention groups. Bayesian negative binomial random intercept models were used to do this (see **Appendix 12** for more detail). This type of model

was chosen because it can adjust for individual characteristics and to allow for variation between police forces.

The terms included in the models were:

- whether each case was intervention or comparison
- protection order
- number of incidents in previous 12 months (for the perpetrator)
- whether the perpetrator had course of conduct offences (for example, harassment, stalking, coercive or controlling behaviours) in the previous 12 months
- whether the perpetrator had violent or sexual offences in the previous 12 months
- a random effect for police force, to allow for regional variations

Limitations of using police data

All forces experienced difficulties and delays accessing data about protection orders and bail conditions. This was not readily available on the crime recording systems that the force analysts had access to, so searching individual case files was required. The delays in accessing this data meant that one force was unable to provide this data in time for inclusion in the outcome evaluation.

There was a substantial amount of missing or inconsistent data provided by the police. For example, victim ethnicity, suspect ethnicity and force-level monitoring data was not always completed in full. Data about the perpetrator warning being given was not always updated. Only 60% of perpetrators were recorded as having been given the warning. In some cases, the warning may have been given at a later date.

Limitations of the approach to the analysis

The start of the intervention was delayed by several months due to the time taken to:

- ensure force engagement
- get data-sharing agreements in place
- pause the intervention to enable comparison group data to be collected

One force was delayed in restarting the intervention after the pause. This meant that the follow-up period had to be reduced to six months rather than 12 months as originally planned. As a result, there was less time to look at longer-term outcomes relating to reoffending and perpetrators being brought to justice.

There were a smaller number of cases included in the intervention group than originally anticipated, which affected the analysis. This was in part due to the forces providing forensic marking products to victims of non-intimate partner domestic abuse, as well as delays in starting. One force only started implementing the intervention six weeks before the end of the trial period, and so fewer victims received the intervention than anticipated. The original plan for the analysis was to use a Bayesian hierarchical model. This would have allowed a nested structure (for example, modelling convictions within the pool of people with follow-up incidents). The small number of forces made this model infeasible, while the small number of activations of forensic marking equipment made the nested structure unnecessary. Instead, a Bayesian negative binomial model with random intercepts was used.

2.1.2. Victim survey

A longitudinal survey was conducted with victims who received a forensic marking device. Five surveys were sent at different time points. These were structured to link with the evaluation outcomes (see **Appendix 2** for the victim questionnaire):

- feelings of safety and freedom
- trust, confidence and satisfaction in the police
- experience of re-offending and unwanted contact from the perpetrator.

The first survey asked about how the victim felt before receiving the intervention, then immediately afterwards. The subsequent four surveys asked how the victim felt at that time, and if there had been unwanted contact since completing the previous survey. Questions were also asked about the extent to which having the forensic marking products had affected their response.

A longitudinal survey design was chosen as it allows for changes over time to be examined. Due to the length of time that victims were involved in the study, the surveys were designed to be short and took no more than five minutes to complete.

Participants

Police officers or staff visited the victims to issue a forensic marking device. During the evaluation period, officers and staff provided the victim with a participant information sheet and explained the purpose and their potential involvement in the research. This included offering incentives. Victims were then asked if they agreed for their details to be passed on to the College to take part in the research. The information provided at this stage was the victim's first name, mobile number and force area. A unique ID was given to each victim to link each survey response over the six-month period.

A total of 186 victims signed up to the research and were sent the first survey. Of these, 96 completed the first survey giving a response rate of 51%. The overall attrition rate was 49%, with the final survey having 49 completions (Table 3).

Table 3: Victim survey response rates.

Survey number	**Time period covered from when forensic marking was received**	**Total number completed**	**Response rate (compared to survey 1)**
1	-1 to 0 months	96	100%
2	1 month	66	69%
3	2 months	61	64%
4	3 months	56	58%
5	6 months	49	51%

Participants had to complete the first survey to be sent subsequent surveys. Participants were also not sent subsequent surveys if they had more than one survey completion missing. If participants responded to four out of the five surveys over the six-month period, they received a £40 voucher after the completion of the final survey.

Demographic questions were only asked in the final survey (n = 49), as an optional request. This was to reduce concerns about victim’s fearing they could be identified.

Of the 49 participants who completed all the surveys, most were aged between 25 and 40 (n = 24) or between 41 and 55 (n = 18) years old. A small number were aged under 26 or over 55 years old. Most described their ethnicity as White British (n = 44), with the remaining participants (n = 5) describing themselves as White non-British, Asian/Asian British or Black/Black British. Nearly all were women (n = 48). Of the 47 who answered the question about disability, 18 said they had a disability.

Data collection

Within a week of signing up, victims were sent the first of five surveys. The next three surveys were sent after one, two and three months. The final survey was sent after six months.

Survey links were sent via an SMS text message to victims' mobile phones. Two reminders were also sent via SMS over the following week if they had not yet completed the survey.

Analysis

The analysis of the survey was linked to the outcomes of trust and confidence in the police and feelings of safety and freedom. Given the size of the sample and the absence of a comparison group, only descriptive statistics were produced. These were analysed to show the change in responses from the first survey (immediately before and after receiving the forensic marking) and the final survey (six months later).

The survey linked to the outcome on reduction in domestic violence and abuse, as victims were asked if they had experienced any unwanted contact from the perpetrator. The survey also asked whether victims had reported the incident to the police. This is important in helping to understand levels of under-reporting.

Limitations of the victim survey

The police provided the victim with information about signing up to the victim survey for the evaluation. In one force, there was very little awareness of the survey among officers. This meant they had not been informing victims of the opportunity to take part in the research. Victims who wanted to take part provided their details by scanning a QR code to take them to a survey page. Some police officers could not access this survey to provide victim contact details on their work phones. These

issues meant that some victims who received the intervention may not have been able to participate in the survey or interviews. This may have contributed to the low initial take-up rate for the survey.

There was a high attrition rate over time between surveys, despite having incentives for completing all five surveys, The lower response rates mean that some detail has been removed in the results section to ensure anonymity. This has substantially reduced the level of intersectional analysis possible.

It was not possible to conduct the survey with victims from the comparison group. This is because the comparison group was identified using existing police data several months after the date of the reference incident.

2.1.3. Interviews

A series of interviews were conducted with four groups of stakeholders (see Appendices 6 to 12):

- police officers and staff from all five forces who were involved in the forensic marking intervention
- domestic abuse perpetrators
- domestic abuse victims who had received forensic marking products
- stakeholders from local and national organisations providing support to domestic abuse victims

A total of 87 interviews were conducted with participants from all force areas in the study.

Participants

Police interviews

Researchers visited each police force involved in the project to:

- gather information about how the project implementation plan had been put into practice
- see where the products were kept and how these were distributed
- see how the intervention was being managed

As part of the visit, interviews were conducted in person with officers where possible. Sampling was purposive in nature, whereby officers who had been directly involved in managing or implementing the intervention were invited to take part in an interview. The project lead arranged the interviews to fit with the officers' shift patterns, and officers often had limited time to spend in the interview. The mean length of the interviews was just over 21 minutes. In-person interviews were recorded using a digital data recorder. Where officers were unavailable during the force visit, interviews took place online using Microsoft Teams at a time to suit the interviewee. A total of 37 interviews were conducted, either online between November 2023 and September 2024 or in person between March and September 2024.

Additional information about how the intervention was designed to be implemented, and how this was delivered in practice, was provided by:

- field notes taken from formal and informal meetings
- observations during the force visits

Victim interviews

Everyone receiving the victim survey was invited to take part in an interview to discuss their experience of being given the products in more detail. The invitation to participate in the interview was included in the first survey, soon after they received the forensic marking products and in the final survey, six months after the first survey. Victims who noted on the survey that they had activated the forensic marking products were invited to take part in an additional interview. For anonymity reasons, demographic characteristics were not recorded. A total of 35 interviews were conducted by telephone between October 2023 and November 2024.

Stakeholder interviews

Specialist domestic abuse sector organisations (national and local) were contacted by email to invite participation. The request was sent to a named person where this was known, through prior contact by research team members, through snowball sampling or from the police locally. A total of 15 stakeholder interviews were conducted online or by telephone between November 2023 and February 2024.

In total, 87 interviews were conducted (Table 4), with participants covering all the police force areas that were involved in implementing the forensic marking intervention (Table 5).

Table 4: Interview participants.

Description	Number of interviews
Police	
Detective sergeant, sergeant or above	11
Detective constable, constable or trainee officer	19
Police staff	7
Non-police stakeholders	
National	8
Local	7
Victims	
Initial survey	20
Final survey and activations	15
Total	**87**

Table 5: Areas covered by police and stakeholder interview participants.

Area	Number of interviewees
Durham	15
Hampshire	9
Kent	10
West Yorkshire	8
National or other local	9
Total	**51**

Analysis

The qualitative data was analysed thematically (Braun and Clarke, 2006) by the researchers who had conducted each set of interviews. The researchers used the expected outcomes as the initial codes, then added to these as new codes were identified from the data. The interview data linked to all the outcomes:

- trust and confidence in the police
- feelings of safety and freedom
- reduction in domestic violence and abuse
- repeat perpetrators being brought to justice

The six stages of thematic analysis used were:

- familiarisation of the data
- generating initial codes
- searching for themes
- reviewing themes
- defining and naming themes
- writing up the analysis

For the police interviews, three transcripts were initially coded by two researchers and compared to ensure consistency. A code book was developed to assist with coding the remaining transcripts. Any queries were resolved between the researchers during the coding process. The codes were then reviewed and revised, and key themes were identified from the data.

In many cases, detail has been removed in the results section to ensure anonymity. This has substantially reduced the level of intersectional analysis possible.

In addition to the thematic analysis, a brief case study was produced for each activation case. These case studies focused initially on the background to the case and events leading up to the activation of the forensic marking device. They described the police investigation, along with the outcome of the case, considering the influence of the forensic marking products on the investigation and outcome. Due to the small number of activations over the intervention period, the cases studies have been adapted to maintain anonymity as far as possible.

2.1.4. Case studies

The original evaluation design included conducting a quantitative analysis of criminal justice outcomes for repeat offenders where forensic marking products had been activated. Few forensic marking activations meant there was little opportunity to look at criminal justice outcomes for repeat offenders, which meant the planned analysis for the outcome around bringing repeat offenders to justice could not be completed. Instead, a case study approach was used, looking in detail at each activation. Where possible, interviews with the victim in each case were conducted to understand their perspective. Interviews were also carried out with the investigating officer and other police officers and staff who had been directly involved in the case. Further notes about the case were accessed from recorded crime data.

A brief case study for each of the five activations was written up, covering:

- the background to the case
- details about the incident where the forensic marking products were activated
- the police response
- the outcome

These case studies link to the outcome on bringing repeat perpetrators to justice.

2.1.5. Cost–benefit analysis

Approach used

To calculate the cost–benefit analysis, a similar approach to that used by Karavais and others (2023) was applied. First, the costs of delivering the intervention to the intervention group were calculated, to give an average cost per person. This takes into account the cost of the forensic marking products, as well as the additional time taken by police officers and staff to deliver the intervention.

To calculate the time spent delivering and administering the intervention, it was estimated that it would take approximately two hours to:

- visit the victim and deliver the spray
- deliver the warning to the perpetrator in custody
- complete the associated paperwork

An additional 15 minutes per victim was included to account for the general management of the intervention and follow-up call to the victim. The grease took longer to apply and would have necessitated an extra visit to the victim.

The total cost of delivering the intervention is the sum of the costs of the products and the time taken. From this, the average cost per person can be calculated.

The next stage is to work out the cost saved by the intervention. To do this, the costs of the crimes that have occurred were calculated. The costs were calculated using the estimations of costs of crime by Heeks and others (2018), which covers:

- costs in anticipation of a crime, such as insurance
- costs as a consequence of a crime, such as physical harm, emotional harm and health service costs
- costs in response to a crime by the police
- other criminal justice costs

There are a limited number of crimes covered by this report. It does not include costs of course of conduct offences, such as stalking, harassment, and coercive and controlling behaviour. An alternative approach to calculating the costs and benefits of the intervention would be to use the figure for the unit cost of domestic abuse suggested by Oliver and others (2019). This figure, which takes account of the repeated nature of many domestic abuse crimes, puts the cost of a domestic abuse offence as £34,015.

Linking cost–benefit to domestic abuse reoffending outcomes

To understand the costs saved by the intervention, this involved:

- calculating the costs of the crimes that were recorded for the comparison group during the follow-up period
- applying the reduction factor due to the intervention to calculate the number of crimes that would have occurred, had the intervention been running

The method for calculating the effect of the intervention on reoffending is discussed in the **Police crime and incident data** section. The costs for the reduced figure were then calculated. The difference is the potential saving from having the intervention in place. The benefit–cost ratio is calculated as follows:

Net benefit	=	Costs of crimes with no intervention – costs of crimes with the intervention
Benefit–cost ratio	=	Net benefit / cost of intervention

2.1.6. Perpetrator interviews

In order to gain the views of perpetrators of domestic abuse, two force areas helped to administer a small number of questions via an online video and linked online survey. One area invited perpetrators via offender managers and the other via custody staff. Perpetrators were asked to watch a short video by a member of the research team that:

- explained the research project to them
- told them how they can access a written participant information sheet
- asked them whether they agreed to participate in the research

Informed consent to participate was recorded at the point of the online survey, which perpetrators could complete themselves or via the person showing them the video. The survey was very short in order to increase participation. It asked about:

- how a person might react to finding out about forensic marking
- whether they thought it would make a person change their behaviour
- whether it would have made any difference to themselves personally
- if there was anything else they wanted to tell the research team

In total, there were seven participants. No demographic characteristics were collected to increase anonymity and reduce time required to complete the survey.

Limitations of perpetrator interviews

Due to the method of sampling and access to perpetrators, the interviews were with a small, self-selecting sample who may not have been representative of the larger population. Those interviewed may have given socially desirable responses, since they were asked via a criminal justice practitioner.

3. Findings

3.1. Summary of key findings

- The forensic marking intervention is likely to have deterred further offending. It was associated with a reduction in repeat incidents over a six-month follow-up period for the intervention group, compared to a matched comparison group. While this reduction was not statistically significant, there was encouraging preliminary evidence of a reduction factor of around 0.78 (equating to 22%).
- The benefit–cost ratio was estimated to be between 38.7 and 82.8. This means that for every pound spent, there would be an economic benefit of between £38.70 and £82.80.
- Survey and interview data showed a consistent increase in feelings of safety and freedom for victims who had received the intervention, which was maintained over time. Many victims felt confident to resume activities outside of their homes that they had stopped doing.
- The victim survey and interview data showed an initial increase in trust in the police to respond appropriately, but this was not maintained over time.
- There were only five activations during the six-month follow-up period. While the forensic marking evidence did not form a key part in any of the prosecutions, it was thought to have reduced investigation time and strengthened cases.

This section sets out the findings from the evaluation, considering each outcome in turn, as set out in the **theory of change**:

- First, the impact analysis of police-recorded crime data on the level of reoffending compared to a matched comparison group is presented, along with the cost–benefit analysis.
- Findings are then presented from the survey responses and interview data relating to feelings of safety and freedom, and around trust and confidence in the police.
- This is followed by an analysis of the survey results around the victims' experience of unwanted contact, and the perpetrators' view of the intervention.
- The final section presents the details of the five case studies where the forensic marking devices were activated.

3.2. Domestic violence and abuse

3.2.1. Summary

The analysis of police data on repeat offences provides encouraging evidence that the forensic marking intervention is associated with a reduction in offending, with an incidence rate ratio of around 0.78 although this is not statistically significant. This means that in the six-month follow-up period, for every 10 crimes in the comparison group, there were 7.8 crimes in the intervention group. This equates to a 22% reduction in the number of crimes. The following section explains how the results of the analysis of police-recorded crime data were calculated, comparing intervention with matched comparison group and the benefit–-cost analysis.

3.2.2. Police data on reoffending

Police recorded domestic abuse offences

Figure 1 shows the number of repeat offences committed by the perpetrator against the same victim as the reference incident, based on the follow-up data from three police forces. The majority of both the intervention group perpetrators (53%) and comparison group perpetrators (51%) did not commit further offences within a six-month period (as recorded by the police). Table 6 summarises the number of offences committed by perpetrators in the intervention group and comparison group during the six-month follow-up period.

Figure 1: Number of offences committed in the follow-up period for the intervention group and comparison group.

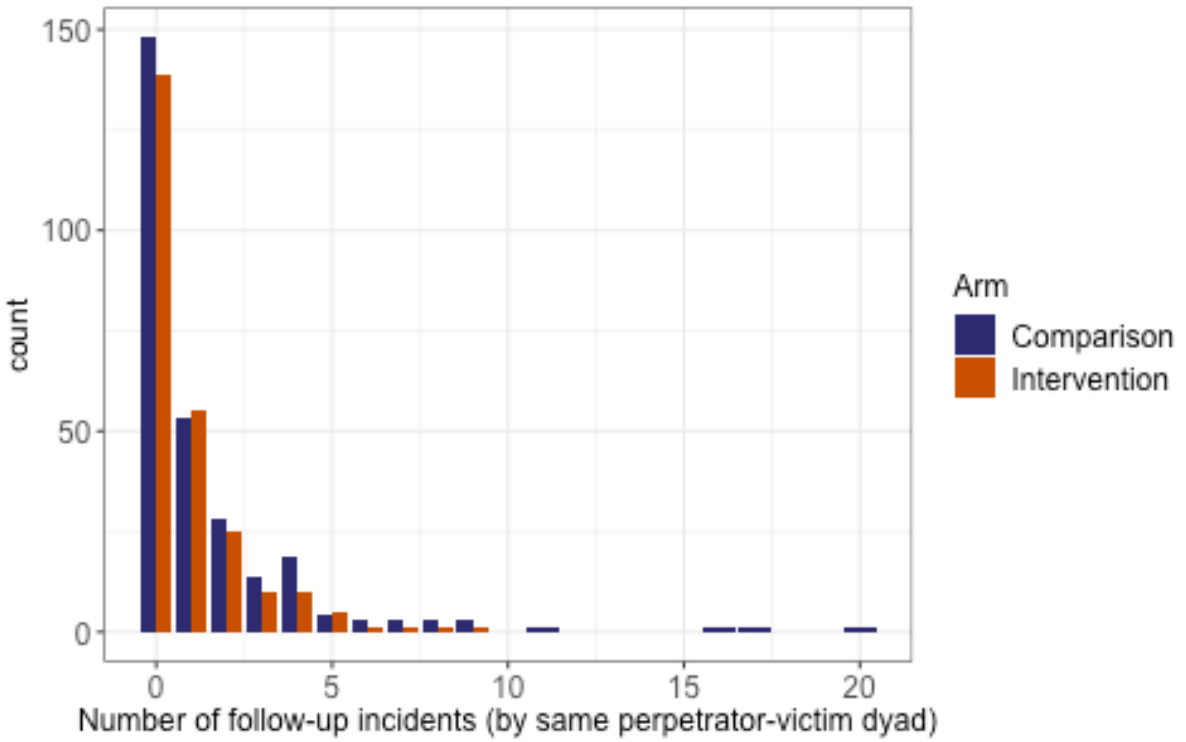


Table 6: Reoffending over six months following the reference incident, for the same victim–perpetrator pair.

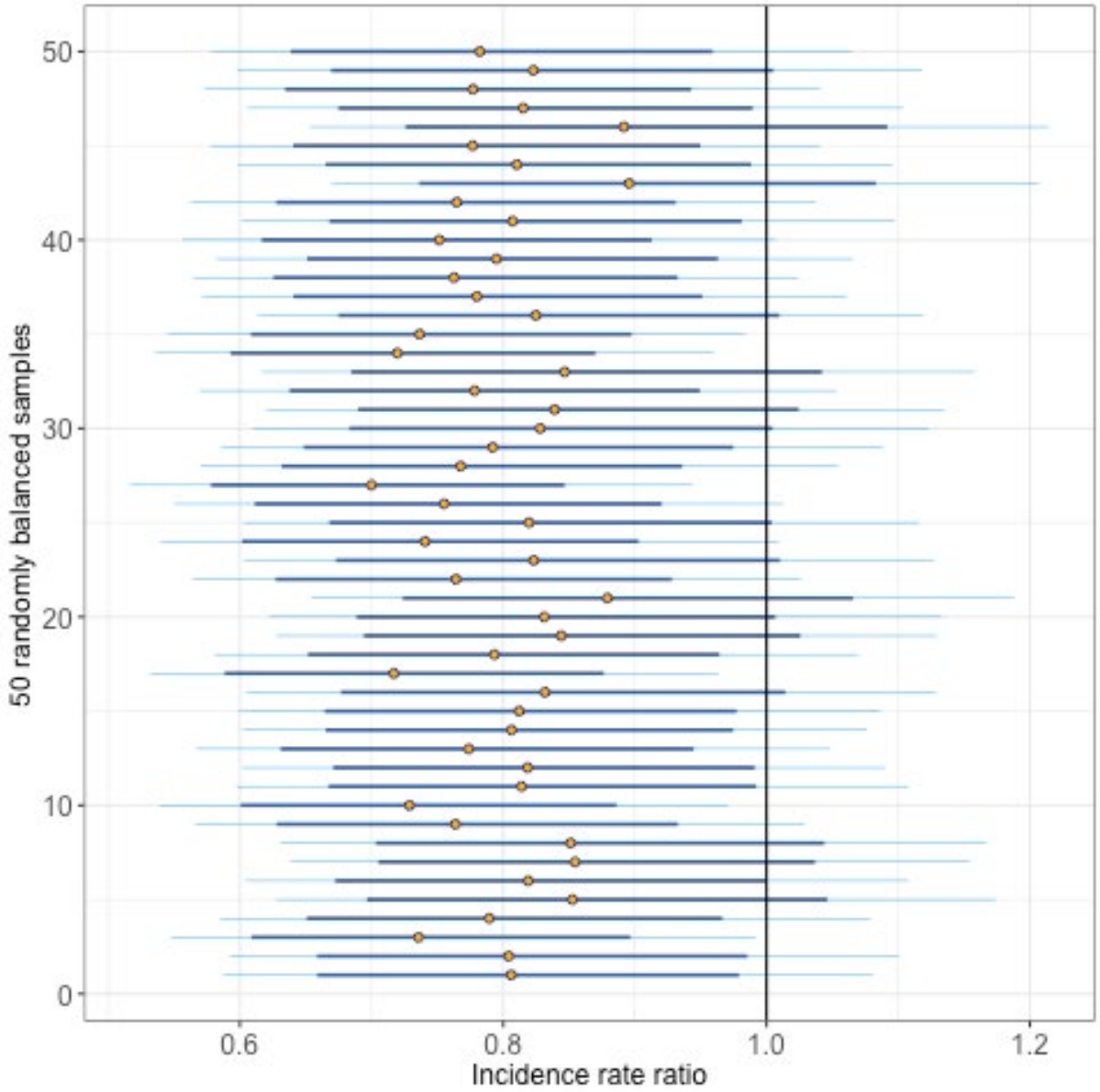
	Intervention group	**Comparison group**
Number of perpetrators	232	276
Perpetrators reoffending within six months	108 (47%)	134 (49%)
Number of offences committed	229	401

The effect of forensic marking on further offending

The police data was then analysed using the model described in the **Police crime and incident data** section. Figure 2 shows the resulting model coefficients for the intervention, indicating that the intervention is associated with a reduction in follow-up incidents for the same victim–perpetrator pair. Each line on Figure 2 represents one randomly sampled data set from the matched intervention and comparison group data, containing n = 378 cases. This was repeated 50 times, with a different

random sample of 378 cases in each data set, which gave a more balanced approach to the matching. For each data set (each line on the y axis), the yellow dot in the middle show the median values of the coefficient. The darker lines show 80% of the probable values for the coefficient (known as the posterior credible intervals) and the lighter lines show the probable values for 95% of the coefficients (95% posterior credible interval). A coefficient value of zero would indicate that the forensic marking intervention has no effect.

Figure 2: Sensitivity analysis of the model to test the effect of forensic marking on further offending.



As Figure 2 shows, all of the median values of incidence rate ratio are below one, indicating that the intervention is associated with a reduction in further offending. The

data indicates that there is encouraging preliminary evidence of a reduction associated with the forensic marking intervention, by a factor of around 0.78. However, for most of the samples, 1.0 is in the 95% credible interval, so these results could not be considered statistically significant.

3.2.3. Benefit–cost ratio

The benefit–cost ratio was estimated to be between 38.7 and 82.8. This means that for every pound spent, there would be an economic benefit of between £38.70 and £82.80. This was based on a reduction of 22% in the number of crimes that would have happened without the intervention in place. This section outlines the calculations for the:

- costs of the intervention and domestic abuse crimes
- benefits of the intervention

Costs of the intervention

Table 7 provides a summary of the costs to consider when implementing the intervention, as well as the potential benefits to victims, the police and the wider criminal justice system.

Table 7: Summary of costs and benefits of the forensic marking intervention.

Costs (inputs)	**Benefits for victims**	**Benefits to police and criminal justice system**
Forensic marking devices	‘Peace of mind’	Less demand due to fewer repeat offences
Officer time	Increased feelings of safety and freedom (long-term, sustained)	Time saved in investigations if the perpetrator admits he was at the scene
Travel costs	Temporary increase in trust and satisfaction with police	
	Reduction in repeat victimisation	

The costs of the intervention were calculated for each of the victims in the three forces that were used for the impact analysis. This included the victims that were given products but not included in the analysis, as they would have contributed to the overall intervention costs. In addition to the cost of the products themselves, the time taken by police officers and staff to deliver the products and manage the overall intervention was calculated. The costs of products provided to the victims in the intervention are shown in Table 8.

Table 8: Costs of forensic marking products.

Product	**Unit cost (£)**	**Total number provided**	**Total cost of products (£)**
Home protection kit	13	74	962
Stickers	3	64	192
Grease	140	7	980
Spray 1	50	108	5,400
Spray 2	45	114	5,130
Mobile unit	1,095	1	1,095
Total		**368**	**13,759**

For police constable hourly rates, West Yorkshire Police use a range from £26 to £34 per hour and suggested an average daily rate of £30 per hour. This average was adopted for the cost–benefit calculations. This is in line with costs used in other studies. For example, Rothwell and others (2022) use a figure of £31.46 per hour, while Gates and others (2024) use a figure of £26.47 per hour.

To calculate the time spent delivering and administering the intervention, it was assumed that it would take approximately two hours to:

- visit the victim to deliver the spray
- deliver the warning to the perpetrator
- complete the associated paperwork

An additional 15 minutes per victim was included to account for the general management of the intervention and follow-up call to the victim. The grease took longer to apply and would have necessitated an extra visit to the victim. The figures for the cost of a 'business as usual' visit to domestic abuse victims for the video first response evaluation (Rothwell and others, 2022) were used to calculate the cost of the journey for this intervention (see Table 9).

Table 9: Costs to deliver the intervention.

Description	Number of hours	Cost per unit (£)	Number of victims	Total (£)
Time to deliver sprays to victim, warning to perpetrator and paperwork	2	30.00	243	14,580.00
Approx additional admin time per victim	0.25	30.00	243	1,822.50
Travel cost per victim (from video first response evaluation)	-	6.16	243	1,496.88
Additional time to apply grease	1	30.00	7	210.00
Additional journey to apply grease	-	6.16	7	43.12
Total				**18,152.50**

The cost of the intervention is product cost + police cost = £13,759 + £18,152.50 = £31,911.50.

The costs of delivering the intervention to 243 victims is therefore approximately **£131.32 per person**.

Costs of domestic abuse crimes

The process to calculate the estimated costs of crimes committed during the intervention was similar to that used by Karavias and others (2023). The costs were

calculated using the estimations of costs of crime by Heeks and others (2018), which covers:

- costs in anticipation of a crime, such as insurance
- costs as a consequence of a crime, such as physical harm, emotional harm and health service costs
- costs in response to a crime by the police
- other criminal justice costs

To maintain consistency with the methods used by Karavias and others (2023), the costs of crimes for the reference offences from the intervention group that have associated costings are included here. This gives an average estimate of the cost of all included crimes (n = 63) as £812,770.00 (Table 10). Dividing this by the number of crimes gives an estimate of the cost of the average crime in the intervention group as £12,901.11.

Table 10: Types, frequency and costs of crimes for the intervention group.

Crime type	Frequency	Crime classification	Unit cost (£)	Weighted cost (£)	Included crimes
Stalking	105	-	-	-	-
Breach	25	-	-	-	-
Actual bodily harm	22	Violence with injury	5,930.00	130,460.00	22
Coercive or controlling behaviour	17	-	-	-	-
Non-fatal strangulation	11	-	-	-	-
Rape	12	Rape	39,360.00	472,320.00	12

Crime type	Frequency	Crime classification	Unit cost (£)	Weighted cost (£)	Included crimes
Criminal damage	9	Criminal damage - other	1,350.00	12,150.00	9
Common assault	7	Violence without injury	5,930.00	41,510.00	7
Grievous bodily harm with intent	7	Violence with injury	14,050.00	98,350.00	7
Harassment	6	-	-	-	-
Threats to kill	6	-	-	-	-
Arson	2	Criminal damage – arson	8,420.00	16,840.00	2
Malicious wounding	2	Violence with injury	14,050.00	28,100.00	2
Kidnap	2	-	-	-	-
Sexual assault	2	Other sexual offences	6,520.00	13,040.00	2
Other	6	-	-	-	-
Missing or not known	12	-	-	-	-
Total	**241**	**-**	**-**	**812,770.00**	**63**
Estimated cost per crime				**12,901.11**	

Adjusting the estimate of the cost of crime for inflation using the Bank of England **inflation calculator** gives a cost per crime (cost #1) of **£15,900.77**.

An alternative approach to calculating the costs and benefits of the intervention would be to use the figure for the unit cost of domestic abuse suggested by Oliver and others (2019). This takes account of the repeated nature of many domestic abuse crimes, and puts the cost of a domestic abuse offence (cost #2) as **£34,015**.

Economic benefits of the intervention

The impact analysis indicated that the forensic marking intervention was likely to be associated with a reduction in offending by a factor of 0.78. To understand the economic benefit, this is applied to the comparison group to look at what the level of reoffending would have been if the victims had received the forensic marking intervention (Table 11). This gives an economic benefit of £1.4 million for cost #1 and £3 million for cost #2.

Table 11: Economic benefit of the forensic marking intervention as applied to the comparison group.

Description	Number of crimes	Cost #1 (Heeks, 2018)	Cost #2 (Oliver, 2019)
Cost per crime		£15,901	£34,015
Comparison group (n = 276)	401	£6,376,209	£13,640,015
Estimated cost if forensic marking had been provided	312.78	£4,973,443	£10,639,212
Economic benefit		**£1,402,766**	**£3,000,803**
Cost of intervention for comparison group	n = 276	£36,244	£36,244
Net benefit		**£1,366,522**	**£2,964,559**
Benefit–cost ratio (economic benefit / cost of intervention)		38.7	82.8

Finally, the cost of the intervention needs to be considered. There were n = 276 victims in the comparison group. The cost of delivering the forensic marking intervention to them would be £131.32 × 276 = £36,244.32.

The net benefit of the forensic marking intervention for the two methods are therefore:

Cost #1: £1,402,765.93 - £36,244.32 = £1,366,521.61

Benefit–cost ratio = economic benefit : cost of intervention

= 1,402,765.93 : 36,244.32

= (1,402,765.93 / 36,244.32) : 1

= **38.7 : 1**

Cost #2: £3,000,803.30 - £36,244.32 = £2,964,558.98

Benefit–cost ratio = 3,000,803.30 : 36,244.32

= (3,000,803.30 / 36,244.32) : 1

= **82.8 : 1**

This analysis gives a range of benefit–cost ratios from 38.7 to 82.8, depending on the figure used for the cost of one crime for this intervention group.

3.3. Safety and freedom

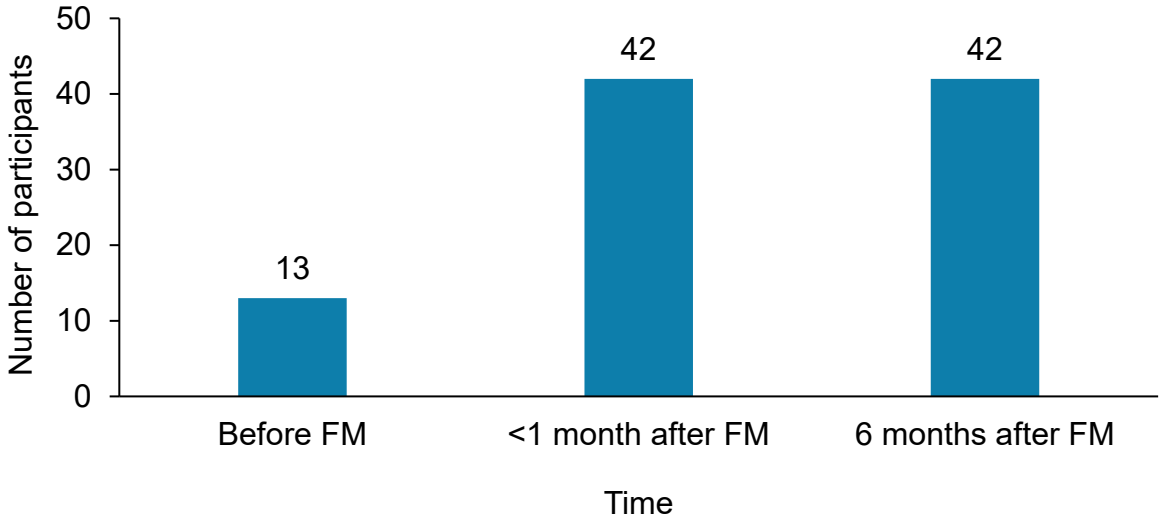
3.3.1. Summary

Both the victim survey and interview data showed a consistent increase in terms of feelings of safety and freedom after receiving a forensic marking device. This increase was maintained over time and was mostly due to the forensic marking device, either “to some extent” or “to a great extent”.

3.3.2. Feeling safe

Figure 3 shows that 13 of the 49 survey participants said they felt safe before being given a forensic device, compared to 42 out of 49 at both one and six months after they were given the device.

Figure 3: Victims who agree or strongly agree with the statement “I feel safe”. FM = forensic marking.



Of the 42 who agreed or strongly agreed that they felt safe after the device (less than one month), 41 said that this was to some extent (n = 19) or a great extent (n = 22) because of the forensic marking. This only dropped slightly for the 42 who agreed or strongly agreed that they felt safe six months on. At this point, 38 still agreed that this was to some extent (n = 25) or a great extent (n = 13) because of the forensic marking.

Most police officers and stakeholders interviewed said they had received positive feedback from victims about increased feelings of safety having received the forensic marking products.

> “I think some [victims] are quite comforted by the fact that they've got something that says, you know, I've got some form of security.”
>
> **– Stakeholder interview**

> “The feedback is amazing. People feel much safer [...] it's made them go about living their normal lives again because they feel they've got that extra protection.”
>
> **– Stakeholder interview**

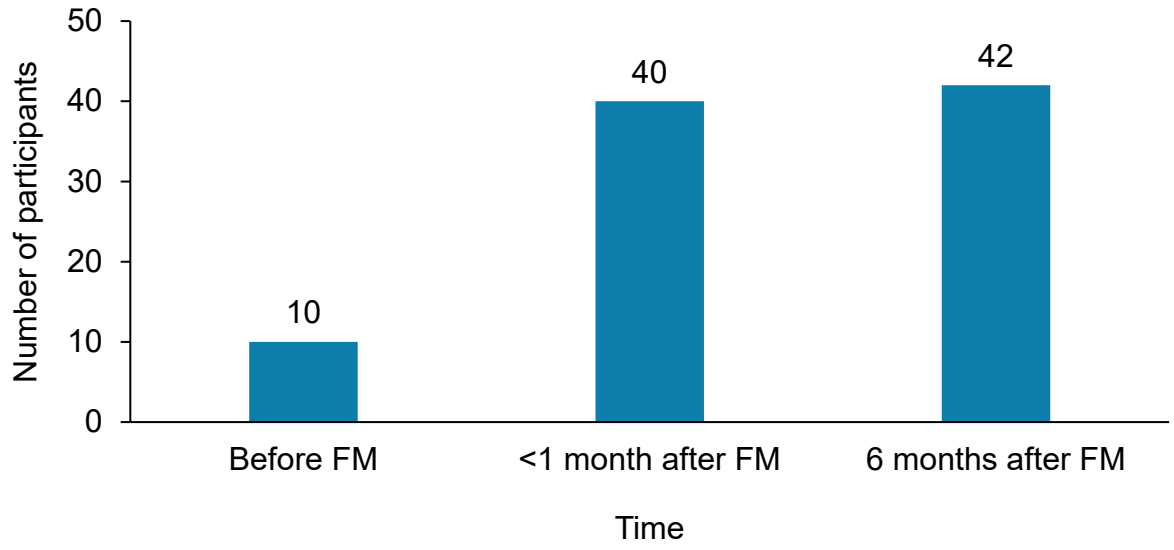
However, there was some concern from around a third of police and stakeholders about:

- the products offering a false sense of security
- victims thinking they need to get close to the perpetrator to activate the spray
- the possibility of using the spray causing an escalation in the level of aggression and assault by the perpetrator

3.3.3. Feeling free to resume daily activities

As understanding has increased about the role that coercive and controlling behaviour plays in relation to domestic abuse, the importance of asking about victim freedom, as well as safety, has become clearer. Figure 4 shows that 10 of the 49 survey participants said they felt free to go about their everyday lives before being given a forensic device. This was compared to 40 at one month and 42 at six months after being given the device.

Figure 4: Victims who agree or strongly agree with the statement "I feel free to go about my everyday life". FM = forensic marking.



There were 40 participants who agreed or strongly agreed that they felt free to go about their everyday life after the device (less than one month). Of these, 39 said that this was to some extent (n = 14) or a great extent (n = 25) because of the forensic marking. This only dropped slightly for the 42 who agreed or strongly agreed that they felt free to go about their everyday life six months on. At this point, 38 still agreed that this was to some extent (n = 25) or a great extent (n = 13) because of the forensic marking.

During interviews, victims talked about feeling safer and freer to go about their everyday lives, echoing the results from the survey. Many talked about feeling able to do things that they had stopped doing. This was particularly the case for those given the spray, as it could be taken out of the house with them, unlike other security measures. Before having the spray, many victims talked about only leaving the house for essential purposes, with some feeling they could not leave the relative safety of their home at all:

> "I literally wouldn't leave the house to walk to the dog before. I'd just let him in the garden, because even in the daytime I feel scared, it's not just in the dark. So yeah, huge benefits. It's a moveable piece of kit that makes me feel more safe wherever I am."
>
> **– Victim interview**

> "I'm so pleased that I still have access to that spray. I've seen my behaviour change in the last year […] I have noticed a real change in the level of my anxiety levels."
>
> **– Victim interview**

Only a small number of victims had returned the forensic marking device by the time of the interview six months on. Whether this was returned voluntarily or not, the impact was felt:

> "I literally carried that for the last year – I carried it everywhere. And yesterday I said I think it's time I handed it in, cause it's silly isn't it, it's just a piece of equipment in your bag but it can make you feel safe."
>
> **– Victim interview**

> "It made me feel a lot safer. And it's now at the point where they've taken the spray from me, and I won't go out. I can't go food shopping. It's completely flipped my head, and I can't get back to the place where I feel physically safe to go out. They took it off me eight weeks ago. I go to school and back now and that's it. At the minute, I've got a court case going on and it's in the area where he lives, and the advice was why don't I get a

> train and then walk. I said there's nothing to stop him coming up to me and my kids. They just said you don't need it because there's been no contact."
>
> **– Victim interview**

3.4. Trust, satisfaction and confidence in policing

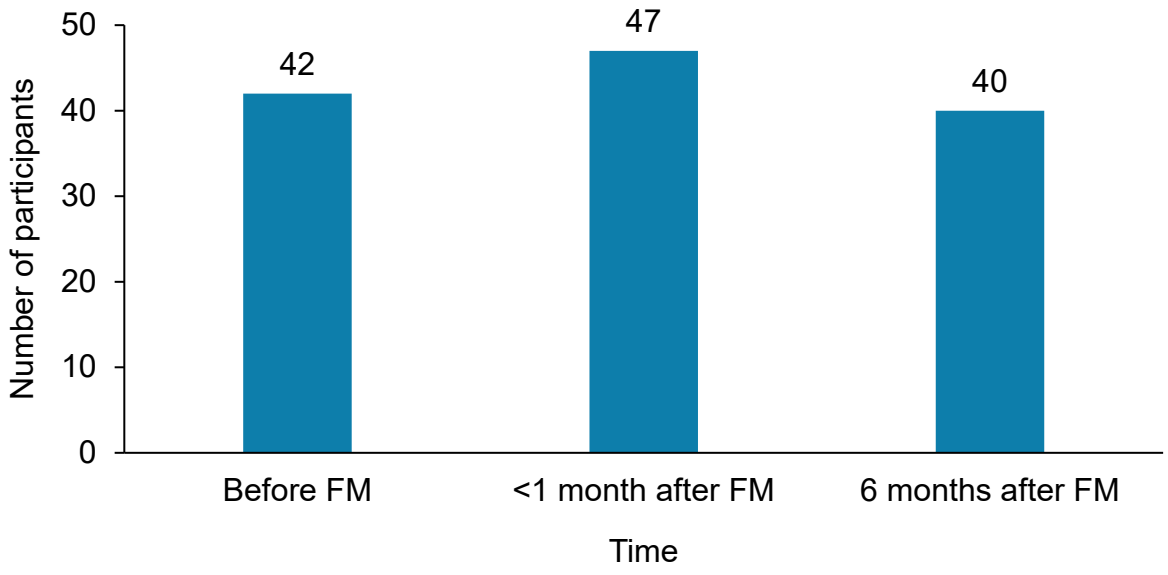
3.4.1. Summary

Both the victim survey and interview data showed a slight initial increase in terms of trusting the police to respond appropriately, but this was not maintained over time. Satisfaction with response and victims' confidence that they would report future incidents also reduced over time.

3.4.2. Trust in the police to respond appropriately

Figure 5 shows there was an initial increase from 42 (before forensic marking) to 47 (after forensic marking) participants agreeing or strongly agreeing that they trust the police to respond appropriately. However, by six months on, this had reduced to below its starting point, to 40 participants agreeing or strongly agreeing that they trust the police to respond appropriately.

Figure 5: Victims who agree or strongly agree that they trust the police to respond appropriately. FM = forensic marking.



3.4.3. Satisfaction with the police response

Satisfaction with the police response followed a similar pattern. One month after the incident for which they were allocated the forensic marking device, 45 victims said they agreed or strongly agreed that they trusted the police to appropriately respond. Of these, 41 said it was due to having the forensic marking to some extent (n = 20) or a great extent (n = 21). Slightly fewer victims (n = 40) said they agreed or strongly agreed that they trusted the police to respond appropriately six months on. Having the forensic marking had less of an influence on their response, with 32 of the 40 victims saying it was due to the forensic marking to some extent (n = 22) or to a great extent (n = 10).

Victims were asked if they felt confident that they would report further incidents to the police. Six months on, victims were less confident in reporting (42 agreed or strongly agreed) than they were after the first month (47 agreed or strongly agreed). A number of victims who previously said they strongly agreed changed their response to ‘agree’.

The survey findings were echoed in the victim interviews. Some victims had felt initially positive about the police in terms of trust, satisfaction and confidence, of which the forensic marking had played a role in this positivity.

> “Yeah, it did [increase trust], because I knew they were taking it seriously, I knew that they weren’t just saying, ‘Oh, it’s another domestic’. I knew that the police were concerned for my safety. That outsiders can see what insiders can’t.”
>
> **– Victim interview**

3.4.4. Peace of mind

Victims frequently used the term ‘peace of mind’ in relation to the questions about trust, satisfaction and confidence in policing, describing how they felt they had been given the tool they needed to prove domestic abuse:

> “Obviously if he denied that he was there it would show that he had been there. So yeah, that’s it, peace of mind.”
>
> **– Victim interview**

> "He always gets away with hitting me and now I can be, 'Yeah, you have hit me and you can't get away with it anymore'. I want to have power over myself if that makes sense. I don't want it for power over him, just over myself."
>
> **– Victim interview**

3.4.5. Sustaining changes over time

The trust, satisfaction and confidence reported by victims initially increased, but this was not sustained over time. In some cases, victims complained about other parts of the criminal justice and family justice systems. This fed into a lack of confidence in reporting to the police, unrelated to the forensic marking. In response to the question, "What would you say to a friend who was offered a forensic marking device?", one victim told us:

> "I'd tell them to take the spray. But my friends, with the police who have been to court, they've said it's a joke, they wouldn't ring them [...] it's serious stuff and it doesn't make you feel safe as a female. And you look on the news and you see they are letting these monsters out and they are tracking their partners down and they're killing them and I don't understand how many more women have to die. You speak out and what happens? It hasn't made my life better. The only thing that's gonna help me is to know that the police are gonna do their job."
>
> **– Victim interview**

Views on the forensic marking devices stayed relatively positive over time. However, the wider challenges with policing and the criminal justice response were too substantial for the forensic marking device alone to maintain victim trust, satisfaction and confidence. While ratings did reduce over time on these measures, it is possible that without forensic marking these would have reduced even further. There is no point of comparison for this data.

3.5. Deterrence from further offending

3.5.1. Summary

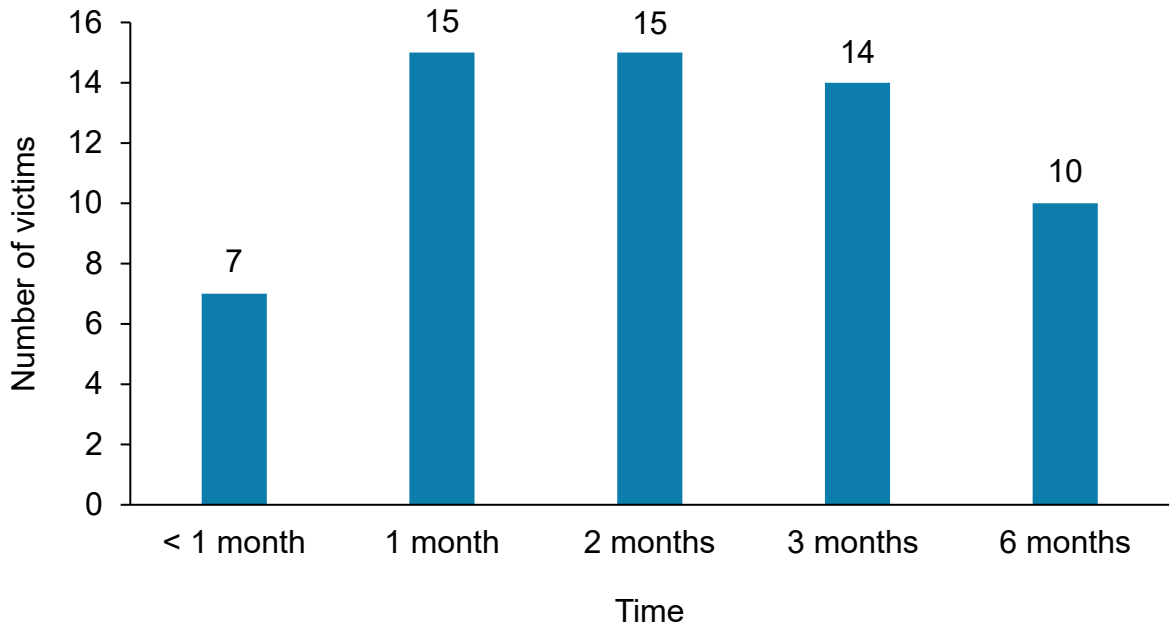
Deterrence was expected to be a key mechanism in the effectiveness of the intervention, with the warning being delivered to perpetrators aimed at deterring further offending. Over half of the victims (27 out of 49) said they had received unwanted contact from the perpetrator, most often through another person. Victims and police officers thought the deterrence effect would work for some perpetrators but not others, and this was echoed in the interviews with perpetrators.

This section presents the findings from the victim survey on experience of unwanted contact.

3.5.2. Victim experience of unwanted contact

Looking again at possible deterrence, the victim survey provides descriptive statistics about the victims' experience of any other unwanted contact by the perpetrator. The victim survey asked about the broader measure of unwanted contact, rather than police-reported incidents or criminal offences. Figure 6 shows that at some point, 15 victims had at least one form of unwanted contact in the six months they were receiving the survey.

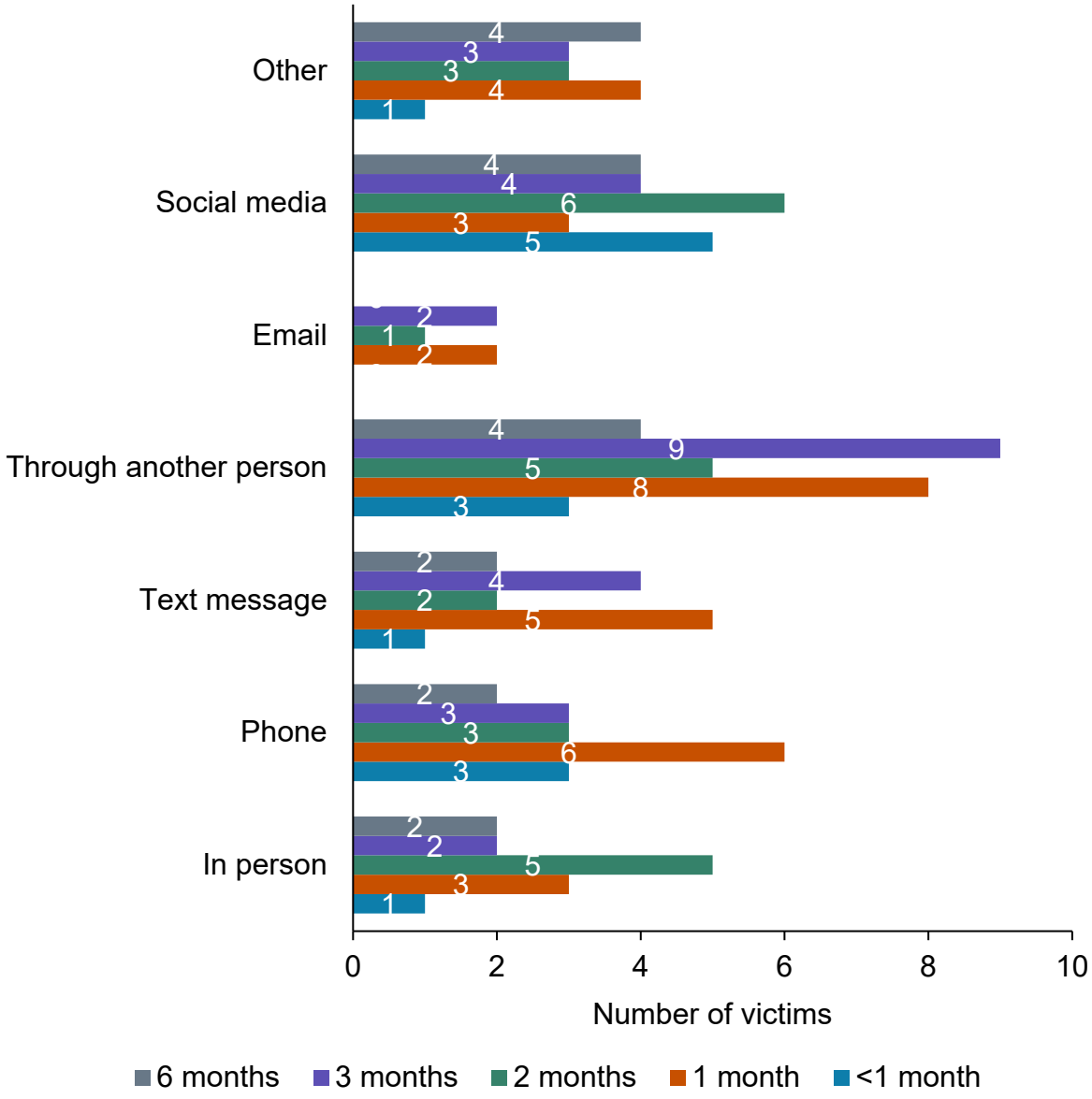
Figure 6: Number of victims with unwanted contacted from the perpetrator.



Due to the surveys being linked over the six-month period, 22 of the 49 victims did not have any further unwanted contact at all. Given there was no opportunity to develop a comparison group for the self-selecting victim survey, it is not possible to compare unwanted contact with those not given the intervention.

For those who said they experienced unwanted contact, we asked about the nature and extent of this unwanted contact and whether they reported it to the police. The nature of the unwanted contact is shown in Figure 7. Note that the numbers exceed victim numbers as more than one form of contact could be selected. Figure 7 shows that 'through another person' was consistently the highest method of contact, and 'email' was the lowest. 'In person' unwanted contact continued for some victims.

Figure 7: Nature of the unwanted contact (n = 49).

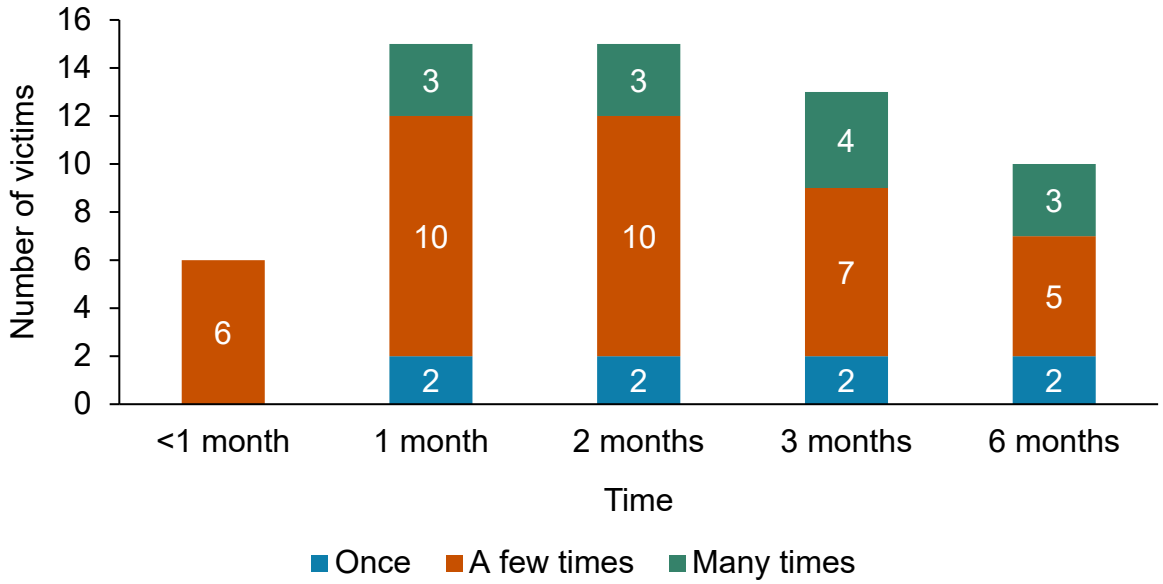


Those participants that ticked the 'other' box were asked to briefly describe the nature of the unwanted contact. The most frequent form was through legal systems (for example, solicitors, Talking Parents App, the family courts or making false reports to the police), which were mentioned a total of eight times. Other forms (in descending order) were:

- being around the victim's home, family, social or work (n = 4)
- economic or financial abuse (n = 2)
- through a shared photo album (n = 1)
- directly contacting victim's employer (n = 1)
- through a letter (n = 1)
- signing the victim up to emails from a pornographic website (n = 1)

Figure 8 shows the extent of the unwanted contact. 'A few times' was defined in the survey as two to four times, and 'many times' was defined as five or more. The graph shows that 'a few times' was the most frequent across all the surveys. Where column numbers add up to lower than they do in Figure 7, this is due to missing data.

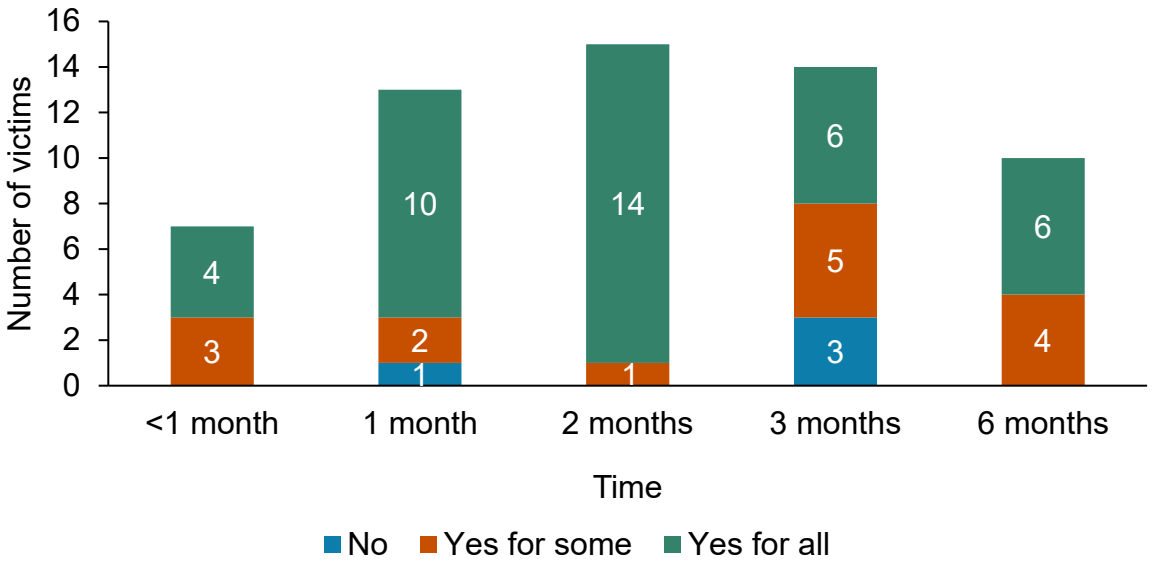
Figure 8: Extent of unwanted contact.



The victim survey asked those who said they had experienced further unwanted contact whether they had reported this to the police. As shown in Figure 9, there was a high proportion of unwanted contacts reported to the police, especially given the lower threshold for these incidents. It is possible that the recorded crime data reflects

increased levels of reporting to the police by the intervention group after being given forensic marking, even if the product has not been activated. This may be because of the initial increase in trust and confidence in the police.

Figure 9: Reporting unwanted contact to the police (n = 49).



The victims interviewed varied on their experiences of further unwanted contact and to what extent they thought the forensic marking (and associated warning) was linked to this. In some cases, there was a cautious optimism that the perpetrator felt they could no longer 'get away' with further contact and bail or order breaches.

However, victims remained unconvinced in a small number of cases, given the failure of previous interventions:

> "He's one of these he's not scared to break the rules. I got a restraining order against him and he rang me from the back of a police car when he got arrested for breaking it. He's just talking to me from the police car! He's got no care at all."
>
> **– Victim interview**

Some victims talked about the ways in which unwanted contact had not stopped but had instead changed. This echoed the survey findings regarding the broad range of ways in which unwanted contact was happening.

> "I think he's found new ways. Now it's not so obvious – the messages are coming through via the family courts, friends and

> family. Like they say he's circling around the court order, he's floating around it, manipulating every loophole, he's not committing a crime, but he is."
>
> **– Victim interview**

Some victims who continued to have unwanted contact talked about feeling let down by the police in ways that were not directly linked to the forensic marking. This was often in relation to civil orders and the police response to breaches.

Around half of the officers interviewed (n = 19) considered deterrence to be a key factor in how the intervention would be most effective. This was through issuing a warning to the perpetrator that the victim has been provided with a forensic marking device. As well as potentially linking the perpetrator to a location they have been told to keep away from, forensic marking evidence could also be used in an evidence-led prosecution. This does not require the victim to engage with the prosecution. A number of officers described the effect on cases where they had provided forensic marking devices and seen a reduction in offending afterwards.

3.5.3. Perpetrator opinions of the deterrence factor

Seven perpetrators participated in the research. Of these, six said that a perpetrator receiving a letter or text from the police saying that forensic devices are in place would have some form of reaction. One participant said they would feel "ashamed" and another said they would feel "concerned". Four participants mentioned feeling deterred:

> "I would think twice because I would not want to go back to prison."
>
> **– Perpetrator interview**

> "Yes because I would not want to be arrested and I know what SmartWater is."
>
> **– Perpetrator interview**

Only one replied that they did not think it would have any impact, saying, "I don't think they would be bothered, and it would not stop them".

Responses were more mixed when asked specifically about whether they thought a person would change their behaviour if they knew the victim had a spray. Some were undecided or felt that it was very individual-dependent, giving responses such as "possibly", "depends on the person", "shouldn't need a spray to change behaviour" and:

> "Not sure, on the TV it doesn't seem to change their behaviours. It doesn't seem to matter what measures are in place."
>
> **– Perpetrator interview**

Two of the seven did feel that it would change a person's behaviour, because of the evidence it would leave, with one stating:

> "Definitely, because there would be definite evidence to convict. It would be like leaving DNA."
>
> **– Perpetrator interview**

Similar responses were given when participants were asked the same question but about the gel or grease option.

Participants were also asked whether they thought it would have made any difference to themselves personally. Four participants said it would – for example, "it would make me stay away for sure", or "yeah I would leave her alone, but it wouldn't get that far". One replied that it would not have made any difference and the other two felt it was not relevant to their situation. This was either because they lived at the same address at the time of the offence or because they had not attended an address that they were not meant to.

These findings are limited in depth and scale. However, they provide additional support to the overall research findings, particularly the deterrent effect in terms of strength of evidence. This included that:

- it is not suitable for all domestic abuse contexts
- while it would have an impact on some perpetrators, there would be others who would not be deterred by this

3.6. Repeat perpetrators brought to justice

3.6.1. Summary

This section looks at the use of forensic marking evidence throughout the criminal justice process, from the response from the police, through to court hearing. It focuses on the cases where the forensic marking products had been activated. Officers had differing views about the main purpose of the forensic marking products. Many officers saw the main aim as providing reassurance to the victim. Several other officers believed that the use of the forensic marking to provide additional evidence should be the focus. As such, officers' main considerations also differed when deciding who should receive the products.

There were very few activations of the forensic marking products during the evaluation period, with only five cases reported to the police across all pilot forces. These are discussed in more detail in the next section. The case studies were informed by:

- interviews with officers who had been involved in the case in some capacity
- interviews with victims
- additional case information from observations in court, media sources and police data

3.6.2. Use of forensic marking evidence in investigations

The evidence from the forensic marking activations did not form a key part in any of the prosecutions that followed the activation incidents. In each case, the perpetrator was arrested at – or close to – the property they were meant to be staying away from, or they admitted they had been at the property. The forensic marking evidence did not form a key part of the cases prepared for court. In one case, it did influence the decision to remand the perpetrator and not release him on bail.

However, the forensic marking element did form part of the investigation process. One officer who had been involved in an activation case said it was helpful in the initial investigation. For each activation case, the perpetrator was aware that the forensic marking spray had been used. The perpetrator was also aware that the liquid showed up on their skin or clothes under UV lights in the custody suite. Officers conducting the interviews thought that the forensic marking evidence helped

with the progression of the interview and potentially speeded up the process. As there was evidence linking the perpetrator to the scene, there was little point in denying they had been there. One officer explained:

> "I do think it's helped him admit that he was present, so I think [it] has had some influence on the outcome overall then."
>
> **– Police interview**

> "It can be the decider between whether or not the perpetrator admits to being there or doesn't admit to being there, and it's particularly helpful with some cases."
>
> **– Police interview**

Another officer believed the forensic marking had strengthened the case. The potential for saving time was discussed by the officers involved. This included reducing the investigation time taken to prove someone has been at a property and preparing the case paperwork for the Crown Prosecution Service (CPS):

> "That saves us an unbelievable amount of time because preparing a not guilty anticipated plea case to a guilty plea case is completely different, the paperwork […] that involves now a lot more paperwork for us. It's redacting incident logs and just the work required is just massive."
>
> **– Police interview**

There was one case where the responding officers were aware that forensic marking had been used and recorded this, but it took some time before the team managing the intervention became aware of the activation.

3.7. Case studies

During the trial period, there were five cases where the forensic marking products were activated and reported to the police. This section explains:

- the background to the case
- why the victim was given the forensic marking products
- details of the circumstances leading to the activation
- the outcome

3.7.1. Case study 1

Overview

The victim deployed the forensic marking when the perpetrator turned up at their home. The police found forensic marking on the perpetrator's jacket. The forensic marking was not used as part of the investigation, as the perpetrator admitted to being present in the interview.

Full case study

The victim had reported an incident to the police where she had a verbal argument with the perpetrator. This argument escalated to physical assault, rape and strangulation, which took place over a couple of days when the perpetrator was at her address. The victim then received forensic marking after this incident, but the perpetrator did not receive a warning letter to say that the victim had received forensic marking.

For the activation incident, the perpetrator turned up at the victim’s address and the victim said she was going to deploy forensic marking. The perpetrator turned away and did not try to gain access to the property and was forensically marked on his back. The police reported that the victim strongly believed he would have tried to gain access and would have seriously assaulted her.

The police were in contact with the victim immediately and deployed another forensic marking canister to the victim the next day. During the investigation, it was alleged that the victim had been contacting the perpetrator and encouraging him “to come round”. The officer conducting the interview said the perpetrator provided “a valid excuse”, which was backed up by further investigation of evidence from their phones. The police confirmed they found forensic marking on the perpetrator’s jacket. However, as the perpetrator admitted to being present in the interview, the forensic marking was not used as part of the investigation.

3.7.2. Case study 2

Overview

The perpetrator breached the restraining order in place and was arrested inside the victim’s home. Forensic marking was found on the jacket of the perpetrator, who

later admitted to two breaches of the restraining order and was given a 46-week custodial sentence.

Full case study

The police were already aware of the perpetrator's previous offences against the victim for stalking, for which he was given a custodial sentence. Forensic grease was applied around the outside of the victim's property. For this activation, the perpetrator was released from prison and went to the victim's home address. The victim was not inside, but a neighbour saw the perpetrator "trying her windows", ringing the bell. The neighbour recorded the perpetrator on their phone.

The following week, the police were trying to locate the perpetrator. During a joint visit to the victim's address with a social worker, they strongly believed he was inside the address. Initially, the victim did not allow entry, but one of the officers saw the perpetrator inside the address. At this point, the officers went into the house and arrested the perpetrator. When in custody under the UV light, the perpetrator's jacket showed evidence of forensic grease, which had been deployed on the windowsills. The police took swabs and sent this off to the labs to confirm that it was the same forensic marking grease that was deployed at the address.

The police reported that the victim was very positive around receiving forensic marking but was reluctant to engage further with the police during investigation.

The perpetrator was aware that forensic marking had been used and responded "no comment" during the interview. According to police, he "did not seem too bothered" about being caught at the address or with the forensic marking on his clothes. He was charged with two breaches of the restraining order and held on remand. He later admitted these offences in court and was given a 46-week custodial sentence.

3.7.3. Case study 3

Overview

The perpetrator turned up at the victim's address and seriously assaulted the victim. The victim deployed the forensic marking during the assault. The perpetrator was arrested and the UV light showed that he was forensically marked. The perpetrator pleaded guilty to actual bodily harm and was given a 38-month custodial sentence.

Full case study

The police were called to attend a domestic incident between the victim and perpetrator at the victim's address. In breach of bail conditions imposed after a previous incident, the perpetrator had turned up at the address intoxicated and had assaulted the victim. During the assault, the victim was beaten and strangled. Fearing for her life, the victim managed to use the forensic marking spray and forensically marked the perpetrator. It was later confirmed that the level of injury was grievous bodily harm. The perpetrator was arrested near the victim's address and taken into custody, where the UV light demonstrated that he had been forensically marked across his chest.

A photo was taken by officers and uploaded to the case file. In interview, the perpetrator did not dispute being at the address. Officers thought this was because the perpetrator knew the forensic spray had been used and would have provided evidence of this. The police reported that even though the perpetrator admitted to being at the address, the forensic marking still added an extra layer of security. For example, the perpetrator could have wanted to later change their statement to not being at the address. The perpetrator later pleaded guilty to a charge of actual bodily harm and was given a 38-month custodial sentence.

3.7.4. Case study 4

Overview

The victim ended the relationship with the perpetrator but the perpetrator continued to make contact with the victim. The perpetrator climbed through the victim's window and she deployed the forensic marking spray. The perpetrator admitted to being present at the victim's address in the interview, so the forensic marking did not form part of the evidence.

Full case study

The victim had previously reported domestic abuse incidents to the police but had later retracted her statements. This time, the victim decided to end the relationship and wanted to pursue charges after being raped by the perpetrator. The victim then reported 10 years of domestic abuse and reported that the perpetrator continued to contact her and come round to her property. The perpetrator did not accept that the

relationship was over. The police provided the victim with forensic marking, alongside other safeguarding measures. However, they had been unable to locate the perpetrator to issue him with the warning letter that states the victim had been provided with forensic marking.

The activation incident occurred shortly after the victim received the spray. During this incident, the victim reported that the perpetrator climbed through the kitchen window and appeared in her bedroom. The victim had the forensic marking spray in her dressing gown pocket, sprayed him and then said it was just water. The perpetrator was arrested and reported that the victim had invited him round. He entered through the front door and said that the victim had asked him to “smell” her new perfume (the forensic marking spray) and was subsequently sprayed with it. During the investigation, the police recovered CCTV of the perpetrator climbing through the window, which supports the victim’s account. Despite not receiving the warning, the perpetrator was aware that he had been forensically marked and referred to SmartWater when having pictures taken of him under the UV light. The perpetrator realised this would provide evidence that he had been at the address. In the interview, the perpetrator admitted he was at the property and therefore the forensic marking did not form part of the evidence.

Police reported that the victim was very positive about the forensic marking as it meant people believed her. The perpetrator was bailed, and the case is still ongoing. Due to the complexity of the case, due to the domestic abuse stretching over 10 years, it has not yet gone to court.

3.7.5. Case study 5

Overview

The victim was issued with a forensic marking device following the perpetrator’s arrest for controlling and coercive behaviour and assault. The perpetrator repeatedly breached bail conditions and, when confronted, the victim was able to use the spray. The forensic marking evidence influenced the decision to remand the perpetrator into custody. The perpetrator received a further custodial sentence and lifetime restraining order following the court appearance.

Full case study

The perpetrator was initially arrested for controlling and coercive behaviour, assault and a serial touching offence. Despite a request for remand, the perpetrator was bailed but breached his bail conditions when he attended the victim's street, where she took a photo of him. The perpetrator was arrested for stalking and again released on bail. The perpetrator had previously assaulted the victim, kicked them out of the house and continued to harass the victim, which was in breach of their bail conditions. The police had put the perpetrator on tag, which helped to monitor curfew compliance, and gave the victim a forensic marking device. The perpetrator was issued with a warning that the victim had a forensic marking product.

Soon after, the victim went to the shops after the perpetrator's curfew time. However, the perpetrator was outside the shops and started shouting and swearing at the victim. The victim used the spray, which allowed her time to get in her car and get away. The perpetrator continued to shout abuse at the victim but seemed confused with what had been sprayed on him and tried to wipe it off. The forensic liquid had been sprayed around his neck and on his jumper.

The perpetrator initially denied being present. However, he was shown to be forensically marked under UV light in custody and there was additional evidence to confirm this. The forensic marking evidence was taken into account in the decision to remand the perpetrator until the hearing at the magistrate's court. The perpetrator was remanded for six months and given a further two-month custodial sentence. On release, the perpetrator had licence conditions and was issued a lifetime restraining order, preventing contact with the victim and restricting access to certain locations.

4. Process evaluation: Implementing forensic marking

4.1. Summary of key findings

- The intervention was largely delivered as planned in the four forces during the trial. However, there was some variation within, and between, forces as to how this was managed in practice.
- Guidance varied between forces as to who should be offered the products, and officers used their discretion to decide who would benefit. Criteria for deciding

who should receive forensic marking included crime type (specifically stalking), cooperation from the victim, and the victim and perpetrator not cohabiting.

- Officers noted that the most efficient way to issue the perpetrator a warning was while they were in custody. However, there were inconsistencies in issuing the warning and recording that this had been done. Only 60% of perpetrators were recorded as having received a warning, despite deterrence being a key factor in the theory of change.
- Most victims described the process by which they were given the device in positive terms and felt confident to use the spray.
- Officers felt that the extra time spent with the victim was seen to be worthwhile if they felt reassured after receiving the intervention.

This section sets out:

- findings from the process evaluation
- how the intervention was implemented in each force, and how this differed between and within forces
- the challenges faced with implementing the intervention as intended
- good practice shared between force leads
- how the intervention was set up and managed
- how decisions were made about eligibility to receive forensic marking products and the process of issuing the warning to perpetrators
- how the trial worked in practice, drawing on findings from the force visits, project documents, and victim and police interviews

4.2. Setting up the intervention

4.2.1. Organisation within the force

The forces were all at different stages of setting up and embedding the intervention, with some inconsistency between forces as to how the intervention was implemented in practice. In West Yorkshire and Durham, the intervention was fairly well established and had been rolled out across all areas. In West Yorkshire, each district had autonomy over how the intervention was implemented, to fit within local structures and safeguarding and domestic abuse teams.

Expanding the intervention meant there was less central coordination on how the intervention was implemented. All districts followed the same general approach to the intervention, and there were efforts made to share ideas and good practice. However, it was very much down to individuals to manage the intervention on a local basis, and this was not always consistent between districts. Given the reliance on individuals to deliver the intervention, staff turnover was highlighted by force leads as being problematic. Forces needed to make sure that the relevant officers were aware of the intervention and trained in its delivery.

West Yorkshire Police had an individual tasked with collating weekly distribution figures for the intervention across the force. This ensured that the district leads were in contact with each other to facilitate information sharing. Likewise, the intervention was well established at Durham Constabulary and had been in place for 18 months before the trial started. Durham had followed the lead from West Yorkshire in adopting the intervention. Many of the officers involved in distributing the products were safeguarding officers, who were involved in investigating high-risk domestic abuse cases. However, response officers and domestic abuse innovation officers were also able to provide forensic marking products. A detective chief inspector was responsible for overseeing the intervention across the force and ensuring it was delivered consistently.

Deploying the intervention in Kent Police was mainly done by officers within the proactive Vulnerability Investigation Team. Although it was available force-wide, in practice, it was mainly focused in one area. A detective sergeant was responsible for overseeing the implementation of the intervention in East Kent.

The intervention was introduced in Hampshire and Isle of Wight Constabulary towards the end of the trial period, having experienced delays in setting up the intervention. This included securing funding and dealing with procurement issues. However, this meant that the intervention was well planned, drawing on the experience of the more established sites. It also gave time to roll out training and raise awareness through their domestic abuse champions network. The project was championed by an inspector and chief inspector, with additional support to ensure the project was administered efficiently.

4.2.2. Managing and administering the intervention

In each force, the project was initiated by an officer of inspector rank or higher. This officer had the authority to embed the project within the force and to make sure other officers and staff were aware of the intervention and how to implement it. Having a more senior officer to promote the intervention, secure funding and take responsibility for oversight of the project worked well. A more junior officer or staff member provided support to oversee the day-to-day administration. Senior officer support also helped to get buy-in for delivering it as intended, as did having supervisors who understood the intervention and promoted its use within their teams.

Officers need to know where the forensic marking products are stored, as well as the process for taking and distributing them, along with the relevant pack of paperwork. Supply was sometimes an issue. Officers in three forces mentioned that victims had to wait for some time before the products could be delivered, due to stocks running low. Delays in product supply and organisational changes meant that one force did not give out enough products to victims during the evaluation period. As a result, they were unable to participate in the impact evaluation.

The administration of the intervention across the force took additional time. This was so a record could be maintained of who received the products, to ensure they received the correct response if they were to report another incident. An important part of managing the intervention was factoring in sufficient staff time to administer the project, and to ensure that the paperwork was completed and uploaded. Administrative tasks included:

- setting up relevant flags on police systems
- completing paperwork
- following up with victims
- promoting awareness of the intervention across the force
- setting up and monitoring a dedicated email inbox for officer queries

In one force, each forensic marking deployment was given an operation name and logged with the control room, with a response plan, in case of further callouts. The operation details, which were stored on Microsoft OneNote, contained information about the deployment. This included photos of the site where the grease had been applied, as well as photos of products provided and the unique bar codes.

4.2.3. Raising awareness of forensic marking

Using a named operation for the intervention and having associated branding was a key mechanism for promoting the intervention within two of the forces.

Within Durham Constabulary, the intervention is known as Operation Wicker, with intranet pages and posters displayed around police stations to raise awareness. Branding was also a key factor in developing the Operation Sentry intervention in Hampshire, with a dedicated intranet page set up alongside promotion through the domestic abuse champions network and additional internal communications.

> "I think having the specific branding has been really eye-catching, you know, because we had so many comments put out internally about this."
>
> **– Police interview**

In Kent, the intervention was known as Operation Highland, but did not have the same prominence within the force.

Having an intranet page with all the relevant information and frequently asked questions was helpful to signpost officers to one place to access the information. This was especially useful given the high turnover of staff in some departments. In practice, awareness was raised more frequently through word of mouth and being tasked to give out the products, with the intranet pages being used for reference.

Officers were also made aware of the forensic marking intervention though training and briefings at the start of a shift. For example, in West Yorkshire, the lead in one district created their own simplified guide to share with officers, in response to having to deal with officer queries. In Hampshire, supervisors received training by the force lead and the forensic marking product supplier, which they were then able to share during staff briefings.

4.2.4. Design and use of the forensic marking products

The handheld sprays were given out far more frequently than other products. Officers had a number of concerns about the design of the forensic marking spray canister from one supplier, which they thought looked like a pepper spray.

> "Now the black case made it look like some form of illegal substance of pepper spray or something. There were all sorts of

> misconstrued ideas, and especially around when the victims are going out into town – for example, for a few drinks or going in, there would be door staff – they were being stopped and asked what it was."
>
> **– Police interview**

Victims also thought it looked like a pepper spray, although this was seen as a benefit to some.

Some victims with physical disabilities were worried about the canister's accessibility and having the strength to remove the safety cap and press the spray down. However, the participant who reported the most concerns about accessibility was one of the victims who did need to activate the spray. She reported that it worked as expected, despite her panicking while doing it.

The canisters from one supplier were larger, to ensure there was sufficient product and propellant within the canister, which meant the spray had further reach (8m to 10m, compared with 4m to 5m). Officers reported that victims preferred the portability of the smaller product, making it easier to carry. This was also mentioned by victims themselves when interviewed.

> "It's much smaller now. I'd had it a year ago – it was a great big chunky thing, it's much better now. I can just hold it in my hand, and I sometimes do when I'm feeling scared."
>
> **– Victim interview**

The forensic grease was used less often than the sprays. One force used it more regularly on the outside of properties, in places to facilitate transfer to the perpetrator if they were trying to access a property other than through the door. Opportunities to use the grease on vehicles to provide evidence of tampering were also explored. In one force, the grease supplied was black in colour. It was therefore clearly visible when applied to white paintwork around doors and windows, looking unsightly and meaning that victims did not want it to be used around their properties. However, some of the national specialist domestic abuse sector stakeholders preferred the idea of the grease to the spray. In particular, they felt that the grease had potential to support sanctuary-style schemes and did not carry the same risk of escalation that they felt the spray did.

4.3. Who received the intervention

4.3.1. Characteristics of victims receiving forensic marking

Most victims who were given forensic marking were female, while most of the associated perpetrators were male. The victims were slightly younger than the perpetrators on average, with median ages of 33 years and 35 years respectively. Information about ethnicity was inconsistently recorded, and was missing for all of the perpetrators from one force.

In terms of the products provided, most people received the spray canister. Home protection kits were predominantly given out in West Yorkshire. Grease was applied to 7% of victims' properties, with the majority of these also receiving a spray. In 60% of cases, the perpetrator received a warning that the victim had been provided with forensic marking products. There was no record of this in 11% of cases. Almost 60% of cases were assessed as being high-risk, while approximately one-third were assessed as medium-risk. Slightly less than half of the people who received forensic marking products had been the victim of stalking.

4.3.2. Decision making about who gets the devices

Suitable cases were identified, mainly by officers from domestic abuse investigation teams. In two forces, response officers were also able to refer victims to receive the intervention. Other referral routes were also used, such as via multi-agency risk assessment conference (MARAC), although less frequently.

All forces had some criteria or guidelines for officers deciding who should receive the forensic marking products, although officers were not always aware of this. These differed slightly between forces. Two forces had broad criteria, requiring only:

- a willingness for the victim to work with the police
- that forensic marking should be used to support the use of protection orders or bail conditions

The other forces provided more guidance, specifying domestic abuse and stalking as suitable crime types, with the additional criteria of the couple not living together.

Risk level as a consideration varied across forces. Officers in one force said risk level was not a consideration. Another force focused mainly on high-risk domestic abuse and stalking at any risk level, although there was some flexibility with this.

There was also a concern that the product would not be suitable in cases with a high risk of further violence, due to the risk of escalation of violence if the spray was activated.

However, in one force, officers and staff said they had not been provided with any criteria or guidance when considering whom the products are most suitable for. This may be more problematic where there is a high turnover of staff, to make sure that all officers understand the purpose of the intervention and how it is intended to be deployed.

Police interviewed also felt that victims needed to be engaged with the idea of having this kind of intervention, and would need to have some level of contact with the police. However, this did not necessarily mean that the victim needed to be engaged with the investigation process and a prosecution.

> "I've had quite a lot of contact this weekend from officers saying, 'Is this case suitable for SmartWater?' And I always say if you think it is and the victim wants it, then the answer is yes."
>
> **– Police interview**

Some officers talked about the suitability of the victim to have the products, and whether they could be trusted to use it responsibly.

4.4. Implementing the intervention

4.4.1. Giving out the products to victims

Once the victim had been made aware of the intervention, they would be contacted by an officer to:

- arrange a time to be given the device
- explain how it worked
- explain when it should and should not be used
- complete the paperwork.

This would either be done as part of a planned visit as part of the investigation process, or a separate additional visit.

Almost all victims who were interviewed described the process by which they were given the device in positive terms. Victims praised the police officers for taking the time to explain how it worked, being knowledgeable about the product, and answering any questions.

Overall, 48 out of 49 victims responding to the survey said that they agreed (16) or strongly agreed (32) with the statement “I would be confident using the spray”.

In a small number of cases, victims had:

- not been told, or had not understood, that they could take the spray out of the house with them
- been left to apply the gel or grease themselves
- been told to use the pen in ways it was not designed for

This was acknowledged by officers in one force, who thought that victims needed better information.

> “I don’t think we gave enough information because there were some little things about a tag on it that victims didn’t know they had to pull the tag off and then press down on the button.”
>
> **– Police interview**

Some victims interviewed who received the intervention felt confident about using the spray device. Other victims were worried about how they would react in the moment. A mixed view was the most common, in that they knew what to do but were not sure if they could do it.

A relatively high proportion of participants in the survey (n = 18) reported having a disability or long-term health condition. A small number were worried about having the hand strength required to remove the safety cap (required on one brand) and pressing the spray down. However, when a victim who had a disability needed to use the spray, she reported being able to do this more easily than expected (see the **Design and use of the forensic marking products** section).

Delivering the intervention increased the time that officers and staff spent with victims. Most thought it took between 10 and 30 minutes to explain how the device

worked and complete the paperwork. A number of officers noted that the extra time spent with the victim was worthwhile if they felt reassured having received the intervention. One officer suggested that the additional paperwork required to log when it had been used would deter officers from offering the intervention. The most time-consuming part was applying the grease to the outside of the victim's property. This required working out the best places to apply it, then taking photos to indicate where it had been applied.

> "So it is extra time […] but it's an extra resource and so it's good."
>
> **– Police interview**

Officers deploying the devices and force leads estimated that it took an average of two hours to complete all aspects of deploying the intervention, including:

- providing the forensic marking products to the victim
- completing the paperwork
- issuing the warning to the perpetrator

As part of the intended deterrence effect, victims were offered stickers or posters to put on their doors and windows warning that forensic products were in use. The nature of these varied between forces but they were usually highly visible, in police colours, with some form of warning message on. In interviews, a small number of victims said they found these helpful, while some had put them up because they thought they had to. A large proportion felt too uncomfortable about making their victimisation more visible in their community:

> "Yeah, but I didn't want to put it up. I live on the village green, it would be like, 'Come and look at what's going on!' I don't think it would have been a deterrent anyway."
>
> **– Victim interview**

4.4.2. Issuing the warning to perpetrators

An important part of the intervention is to warn the perpetrator that the victim has received forensic marking products. When activated, these can provide proof that the perpetrator has been in close proximity to the victim. The warning is also designed to deter perpetrators from re-offending. Some officers thought that perpetrators who

had previously breached bail conditions or protection orders were unlikely to pay attention to the warning letter. There were instances where the perpetrator had a history of not being deterred by bail conditions and being likely to reoffend. In such cases, the forensic marking intervention can provide a means of gathering evidence if they continue to breach once the intervention has been deployed. During the intervention, the perpetrator was warned about the victim receiving forensic marking products in around 60% of cases.

The majority of officers noted that the warning was delivered in person to the perpetrator, most often when they were in custody. This was seen to be the most straightforward way of delivering the message about the forensic marking devices being deployed to the victim.

In-person delivery was time-consuming if the warning could not be delivered in custody. Alternative methods of delivery were mentioned, such as phoning the perpetrator. One officer described how a perpetrator had been bailed to stay at an address out of the force area. As it was not practical to deliver the warning in person, the officer emailed the perpetrator's solicitor with details of the warning.

Officers from two forces preferred to send a letter to the perpetrator's address, in one case to avoid facing the perpetrator's reaction. However, many officers noted that the perpetrator did not seem too bothered by being issued with the warning.

There were mixed views from officers on whether they thought issuing the warning was a good idea. Most thought the warning was an effective preventive measure, to deter the perpetrator from further offending. Others thought that it would be better not to tell the perpetrator.

> "I'd rather not give them a letter. I know the whole point is the deterrent effect, but I'd rather they didn't know."
>
> **– Police interview**

> "I think that the warning letter is the thing that stops them attending."
>
> **– Police interview**

In a number of cases, the victim did not want the perpetrator to know about the forensic marking device.

> “So we spoke to her about the SmartWater. But then he would have known, actually, she's spoken to the police to get the SmartWater. So she turned it down.”
>
> **– Stakeholder interview**

Alongside this, some stakeholders had concerns that issuing warning letters to the perpetrator would act as a catalyst for escalation.

Another was concerned that once someone had been sprayed, they might ‘have nothing to lose’ and increase the risk of serious harm:

> “Why would he not beat her up black and blue? He already has been sprayed, he’s already going to prison, why would he not punish her for that?”
>
> **– Stakeholder interview**

In the interviews with victims, mixed practice was reported. Some victims were not informed that the perpetrator would be told they had a forensic marking device and only found out this was a possibility when asked the question in the interview. At the other end of the spectrum, some victims were shown the letter that would be given to the perpetrator.

4.4.3. Re-issuing forensic marking sprays

Victims who received forensic marking products have a follow-up from officers to check if they are happy with the products and want to retain them. In one force, the intention was to follow up with the victims receiving forensic marking products every month. In reality, as the number being distributed increased, this level of contact was no longer sustainable, with one person in each district often being responsible for the follow-up calls. Instead, they aimed to contact victims to follow up with them every three months.

With limited supplies and limited budgets for purchasing forensic marking products, officers in three of the forces spoke about the possibility of reusing sprays that are no longer needed. This was also the case where the perpetrator was convicted and received a custodial sentence.

> “I did actually contact somebody recently. We had quite low supplies, so I’d contacted her just to say, you know, ‘What’s

> happening with you and him? Are you back in a relationship? Do you feel like you still need it?' And she was like, 'I don't need it at all, I do still speak to him, but we're not in a relationship and I don't feel like I would benefit from it anymore' and she returned it."
>
> **– Police interview**

In practice, few of the canisters were returned to the police. Over the time the intervention has been running, Kent collected eight back from people whereby it was determined that they are no longer required. Kent only re-issued one canister to a new victim. In West Yorkshire, around 30 had been returned from over 300 issued. In Durham, three canisters were returned and two were reissued, from almost 200 devices given out. None had been returned in Hampshire, and they did not plan to collect any back from victims.

On a practical note, forces need to consider what happens to the products if the victim moves out of the police force area. Are follow-ups going to continue, in which case for how long? Are products that are no longer needed going to be returned and reissued?

5. Conclusions

A mixed-method, longitudinal impact evaluation was undertaken of a forensic marking intervention for domestic abuse. Most victims received spray canisters, which could be used by the victim to spray the perpetrator in cases of unwanted contact. This could potentially provide evidence that the perpetrator had been in close proximity to the victim.

The data analysis indicates that the forensic marking intervention is associated with a reduction in offending when provided to victims of intimate partner abuse. The intervention was considered relatively straightforward and quick to implement alongside other safeguarding measures, and provided a positive benefit–cost ratio. There were very few activations over the evaluation period, which may indicate that the intervention had a deterrent effect. However, the warning was only given to around two-thirds of perpetrators. For most of the case studies where the forensic marking had been used, the perpetrator had not been given a warning.

The forensic marking intervention was well received by most victims, and was largely supported by the police officers who had been involved in delivering the intervention. These officers considered it as being one of few preventive measures available for victims of domestic abuse. Victims reported substantial increases in feelings of safety and freedom after receiving the products. Victims were also able to resume routine activities outside of the home that they had stopped doing.

There were concerns from a small number of police officers and other stakeholders about the products giving a false sense of security. There was also the possibility that using the spray could cause an escalation in the level of violence, However, there was no evidence of this happening during this trial.

5.1. Safety and freedom

Both the survey and interview data showed a consistent increase in terms of feelings of safety and freedom for victims. This increase was maintained over time and most victims said this was to some extent, or a great extent, due to having the forensic marking device. Many victims felt confident to resume activities outside of their homes that they had stopped doing, as they could take the spray with them.

Victims were warned not to put themselves at risk. However, some officers and stakeholders were concerned that the victim may still put themselves at risk in trying to spray the perpetrator, rather than avoiding interaction with them. They felt that asking victims to get close enough to spray the perpetrator was contradictory to the usual advice, which was to tell the victims not to interact with the perpetrator.

5.2. Trust and confidence in the police

Both the victim survey and interview data showed an initial increase in terms of trusting the police to respond appropriately, but this was not maintained over time. Satisfaction with the police response and confidence that victims would report future incidents also reduced over time. This may reflect a positive reaction to the initial response from the police in being given the forensic marking products, which may not have been sustained as the investigation continued.

5.3. Reduction in domestic violence and abuse

To assess whether there was a difference in the number of further offences between the comparison and intervention groups, Bayesian negative binomial random intercept models were used. There was an association between the forensic marking intervention group and a reduction in incidents over a six-month follow-up period. This was not considered significant, but there was compelling preliminary evidence of a reduction at around the level of 0.78.

In the self-report survey, slightly less than half of the victims did not have any further unwanted contact at all in the six-month follow-up period. When contact did occur, it was mainly though other people, rather than direct contact with the perpetrator. Most victims experiencing unwanted contact continued to report this to the police for some or all of the incidents.

Victims who were interviewed reported on varied experiences of further unwanted contact and to what extent they thought the forensic marking (and associated warning) was linked to this. In some cases, there was a cautious optimism that perpetrator felt they could no longer “get away” with further contact and bail or order breaches. However, in a small number of cases, victims remained unconvinced due to the failure of previous interventions. Some police officers were also sceptical of the warning being adhered to by perpetrators who had previously disregarded bail conditions or protection orders to stay away from the victim. An important part of the intervention was to issue the warning to the perpetrator that the victim has received forensic marking products. However, the perpetrator was only warned about the victim receiving forensic marking products in around 60% of cases.

5.4. Repeat perpetrators brought to justice

One of the aims of the intervention is to provide strong evidence of a perpetrator being close to the victim. However, during the course of the evaluation period, there were very few activations of the forensic marking products. Only five cases were reported to the police during this time across all the police forces involved. The evidence from the forensic marking activations did not form an important part in any of the prosecutions that followed the activation incidents. In each case, the perpetrator was arrested at – or close to – the property they were meant to be staying away from, or they admitted they had been at the property.

However, the forensic marking element did form part of the investigation process, and was thought to have reduced investigation time and strengthening cases. Investigating officers noted that forensic marking evidence can influence the course of an interview. It may lead a perpetrator to admit they were present at a location, rather than denying their presence or giving a no-comment interview.

5.5. Costs and benefits

The cost of the forensic marking products and time taken to deliver the intervention were relatively low. Forensic marking products were often given to the victim during a visit as part of the investigation or to discuss other safeguarding measures. In every force, there was uncertainty around budgets and securing short-term funding for the intervention. This led to concerns about having enough products available to provide to victims. The intervention needs to have good administrative oversight to:

- know how many unused products are available and where these are stored
- keep record where products have been used
- follow up with victims about whether they want to retain products

The average cost of delivering the intervention is estimated to be around £131.32 per person. This varied, as the sprays had lower costs than the grease. Depending on the method used, and assuming a reduction in follow-up incidents of 0.78, the benefit–cost ratio was estimated to be between 38.7 and 82.8. This means for every pound spent, there would be an economic benefit of between £38.70 and £82.80.

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Appendices

Appendix 1: Instructions for police officers for recruiting victims to the survey

Contact details for the forensic marking survey

The College of Policing is evaluating the forensic marking intervention to see how effective forensic marking products are in reducing unwanted contact and supporting detection of crimes. This information sheet provides details of how officers and staff providing the forensic marking products can engage the people receiving the products in the evaluation.

1. **Provide a participant information sheet to the person receiving the forensic marking products.**

   This explains what the evaluation is about and what will happen if they take part. Participants will be asked to complete a short survey about their forensic marking products, sent in a link by text message. The first survey will be sent within a week of receiving the forensic marking kit, with four follow-up surveys over the next few months.

2. **Ask for consent to pass on their contact details to the College research team.**

   If they are interested in taking part in the survey, please ask them to provide consent for you to send their contact details to the research team (first name and mobile phone number, along with the date and police force).

   Providing contact details at this stage does not commit them to having to participate in the evaluation. They can decide whether or not to complete the surveys when they receive them.

3. **Complete the form and enter contact details.**

   Scan the [QR code provided] or go to **https://forms.office.com/e/aZza4MFviG** and complete the contact form. Alternatively, email the research team at **forensic.marking@college.police.uk**

   College researchers will send the link to the survey within one week. The text message will be sent from **Survey Team**.

Appendix 2: Participant information sheet – forensic marking survey (victims)

The College of Policing is the professional body for the police in England and Wales. The College provides those working in policing with the skills and knowledge necessary to prevent crime, protect the public and secure public trust. To enable us to fulfil our purpose, we rely on volunteers to participate in research.

For this project, we are working in partnership with researchers from the Durham University Centre for Research into Violence and Abuse (CRiVA).

We would like to invite you to take part in a research project. This information sheet tells you why the research is being done, what it will involve for you, what information we will ask from you, and what we will do with that information.

We will, in the course of this project, be collecting personal information. Under the General Data Protection Regulation (2016), we are required to provide a justification (what is called a “legal basis”) in order to collect such information. The legal basis for this project is “task carried out in the public interest”.

What is the purpose of the research?

We want to evaluate a forensic marking intervention to see how effective it is. People who have experienced domestic abuse are provided with products by the police (such as a handheld spray or gel), which can forensically mark a perpetrator to provide evidence that they have been present at a specific location. Perpetrators receive a warning that the victim has received a forensic marking kit, to deter them from offending again. The research is funded by the Cabinet Office.

Why have I been invited to take part?

You have received this invitation as you have recently been provided with forensic marking products by the police.

What will happen if I agree to take part?

If you agree to take part, you will be asked to provide your consent at the start of each survey. Even if you agree to take part at the beginning, you can decide not to complete the surveys, and you can opt out of receiving further surveys.

You will be asked to complete five short surveys, sent to you in a link by text message over the next six months. These will ask about your experience of having the forensic marking products, and about satisfaction and trust in the police. The first survey will be sent to you within a week of receiving the forensic marking kit. The next three will be sent at approximately one-month intervals, with the final survey being sent six months after the initial survey.

The survey will be short and only take between 5 and 10 minutes to complete each time.

What are the benefits for me in taking part?

Your involvement will help us to understand if using forensic marking products is an effective way of providing support for people who have experienced domestic abuse and preventing further harm.

If you complete all five surveys, you will receive a £40 voucher to recognise your contribution to the research (you will be given this after completing the final survey).

Are there any risks for me if I take part?

The research is unlikely to pose any risk or disadvantage to you. The questions relate only to the products you received, any unwanted contact with the perpetrator and your experience of dealing with the police. If you did find that this causes you any distress, contact details for support services are listed at the end of the information sheet.

If you change your mind about participating in the research, you can withdraw your data up to seven days after you have completed the survey by sending your request to **forensic.marking@college.police.uk** and citing the last four digits of your mobile phone number.

What will you do with my information?

Any information you provide will be treated as confidential and research participants will be anonymised in any reports of findings from the research, so you will not be identified. The exception to this is where you tell us something that indicates that you or someone else is at risk of harm. In this instance, we may need to share this

information with a relevant authority. However, we would speak to you about this first.

Anonymised personal data will be gathered from police records from all those who agree to take part in the interviews, to provide information about participants' age, gender and the products received.

At the completion of the project, we will retain anonymised data from the research, which will be stored securely on College and Durham University systems for 10 years, in line with College policies.

We will use the data collected in this research to inform policing policy and practice, through the preparation of research reports, presentations and academic outputs, such as books or articles. We will place publicly accessible reports and academic outputs on the College and Durham University websites.

What happens next?

If you agree to take part in the survey, the police will provide the research team with your name and mobile phone number so we can send you the link to the survey by text message. So you can recognise our survey, the text message will be sent to you from **Survey Team**.

If you have any questions, please contact Rosie Erol at **forensic.marking@college.police.uk**

Support organisations

Victim Support: **www.victimsupport.org.uk** or 0808 1689 111

Women's Aid: **www.womensaid.org.uk**

National Domestic Abuse Helpline: 0808 2000 247

Men's Advice Line: 0808 801 0327

Appendix 3: Victim surveys

A longitudinal survey was conducted with victim-survivors who received a forensic marking device. There were five surveys over a six-month period.

Survey 1 was sent out when the victim first signed up, within a month of them receiving the forensic marking products. Surveys 2, 3 and 4 were the same. Each of these surveys were sent out one month after the one before. Survey 5 was sent out three months after survey 4 and asked for additional demographic information.

Victim survey 1

The first survey was sent to victims within one week of them receiving the products and signing up to the survey.

1. Thank you for considering taking part in this survey about forensic marking products (spray, grease and/or gel). Further details of the project can be found in this **information sheet** you were given by the police or by emailing **forensic.marking@college.police.uk**

   This is the first of five surveys we will send to you over the next six months. Each survey will take around 5 minutes to complete. At the end of the fifth survey you will receive a £40 voucher to recognise your contribution to the research.

   I have read and understood the participant information sheet and I agree to take part in this research project

   - Yes
   - No

2. When were you given the forensic marking device?

   - Within the last month
   - Over a month ago

   Thinking back to the month before you received the forensic marking product(s) (spray, grease or gel), how much would you agree with the following statements:

3. I felt safe

   - Strongly agree
   - Agree
   - Disagree

- Strongly disagree

4. I felt free to go about my everyday life
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
5. I trusted the police to respond appropriately
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
6. I feel safe
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
7. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent
   - Not at all
8. I feel free to go about my everyday life
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
9. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent
   - Not at all

10. I trust the police to respond appropriately
    - Strongly agree
    - Agree
    - Disagree
    - Strongly disagree
11. To what extent is this due to the forensic marking product(s)?
    - To a great extent
    - To some extent
    - Not at all
12. I am satisfied with the police response
    - Strongly agree
    - Agree
    - Disagree
    - Strongly disagree
13. To what extent is this due to the forensic marking product(s)?
    - To a great extent
    - To some extent
    - Not at all
14. I would be confident reporting any further incidents to the police
    - Strongly agree
    - Agree
    - Disagree
    - Strongly disagree
15. (If you have the spray) I would be confident using the spray if needed
    - Strongly agree
    - Agree
    - Disagree
    - Strongly disagree
    - I do not have the spray

16. Since receiving the forensic marking product(s) have you had any unwanted contact from the person who has been warned to stay away from you?
    - Yes
    - No
17. How many times have they contacted you?
    - Once
    - A few times (2-4)
    - Many times (5+)
18. How did they contact you? Please tick all that apply
    - In person
    - Phone call
    - Text message
    - Through another person
    - Email
    - Social media
    - Other (please specify):
19. Did you feel able to tell the police about this contact?
    - Yes, for all contacts
    - Yes, for some contacts
    - No, for no contacts
20. Was the forensic marking product(s) activated during any of these contacts?
    - Yes, the spray
    - Yes, the gel / grease
    - No
    - Not sure
21. It's really important that we hear from people who have had the device activated – if you have used the spray (if you have this) or if you suspect the gel (if you have this) has transferred to the person who has been told to stay away from

you. Please add your telephone number or email address if you are happy for us to contact you about this.

22. Is there anything else you'd like to tell us about having the forensic marking product(s)?

    We hope that this research will improve police responses. We are also hoping to speak to some people in more depth by phone. Those who participate will receive a £15 voucher to recognise their additional contribution to the research. For further details please click **here** for the participant information sheet.

23. If you are interested in talking to a researcher in more depth about your experiences with the forensic marking product(s), please leave your mobile number and / or email address for us to contact you with more details.

    Thank you for taking the time to respond, we will contact you again in 1 month.

    To opt out of receiving further surveys for the evaluation please email **forensic.marking@college.police.uk**

Victim survey 2

(Note that the questions here were worded identically for victim surveys 3 and 4.)

Thank you for considering taking part in this survey about forensic marking products (spray, grease and/or gel). Further details of the project can be found in this **information sheet** you were given by the police or by emailing **forensic.marking@college.police.uk**

This is the second of five surveys we will send to you over the next six months. Some of the questions may be the same. Each survey will take around 5 minutes to complete.

At the end of the fifth survey, you will receive a £40 voucher to recognise your contribution to the research.

1. I have read and understood the participant information sheet and I agree to take part in this research project
    - Yes

- No

Thinking about the last month, how much do you agree with the following statements:

2. I feel safe
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
3. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent
   - Not at all
4. I feel free to go about my everyday life
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
5. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent
   - Not at all
6. I trust the police to respond appropriately
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
7. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent

- Not at all

8. I am satisfied with the police response
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
   - I have had no contact with the police in the last month
9. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent
   - Not at all
   - I have had no contact with the police in the last month
10. I would be confident reporting any further incidents to the police
    - Strongly agree
    - Agree
    - Disagree
    - Strongly disagree
11. (If you have the spray) I would be confident using the spray if needed
    - Strongly agree
    - Agree
    - Disagree
    - Strongly disagree
    - I do not have the spray
12. Thinking about the last month, have you had any unwanted contact from the person who has been warned to stay away from you?
    - Yes
    - No
13. How many times have they contacted you?
    - Once

- A few times (2-4)
- Many times (5+)

14. How did they contact you? Please tick all that apply
    - In person
    - Phone call
    - Text message
    - Through another person
    - Email
    - Social media
    - Other (please specify):
15. Did you feel able to tell the police about this contact?
    - Yes, for all contacts
    - Yes, for some contacts
    - No, for no contacts
16. Was the forensic marking product(s) activated during any of these contacts?
    - Yes, the spray
    - Yes, the gel / grease
    - No
    - Not sure
17. It’s really important that we hear from people who have had the device activated – if you have used the spray (if you have this) or if you suspect the gel (if you have this) has transferred to the person who has been told to stay away from you.

    Please add your telephone number or email address if you are happy for us to contact you about this.
18. Is there anything else you'd like to tell us about having the forensic marking product(s)?

    Thank you for taking the time to respond, we will contact you again in 1 month.

To opt out of receiving further surveys for the evaluation please email **forensic.marking@college.police.uk**

Victim survey 5

Thank you for considering taking part in this survey about forensic marking products (spray, grease and/or gel). Further details of the project can be found in this **information sheet** you were given by the police or by emailing **forensic.marking@college.police.uk**

This is the final survey, which will take around 5 minutes to complete. At the end of this survey, you will find out how to get your £40 voucher to recognise your contribution to the research.

1. I have read and understood the participant information sheet and I agree to take part in this research project
   - Yes
   - No
2. Do you still have your forensic marking products?
   - Yes
   - No
   - Other (please specify)

   Now that we are at the end of the survey series, how much do you agree with the following statements:
3. I feel safe
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
4. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent

- Not at all

5. I feel free to go about my everyday life
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
6. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent
   - Not at all
7. I trust the police to respond appropriately
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
8. To what extent is this due to the forensic marking product(s)?
   - To a great extent
   - To some extent
   - Not at all
9. I am satisfied with the police response
   - Strongly agree
   - Agree
   - Disagree
   - Strongly disagree
10. To what extent is this due to the forensic marking product(s)?
    - To a great extent
    - To some extent
    - Not at all
11. I would be confident reporting any further incidents to the police

- Strongly agree
- Agree
- Disagree
- Strongly disagree

12. (If you have the spray) I would be confident using the spray if needed
    - Strongly agree
    - Agree
    - Disagree
    - Strongly disagree
    - I do not have the spray
13. Since the last survey, have you had any unwanted contact from the person who has been warned to stay away from you?
    - Yes
    - No
14. How many times have they contacted you?
    - Once
    - A few times (2-4)
    - Many times (5+)
15. How did they contact you? Please tick all that apply
    - In person
    - Phone call
    - Text message
    - Through another person
    - Email
    - Social media
    - Other (please specify):
16. Did you feel able to tell the police about this contact?
    - Yes, for all contacts
    - Yes, for some contacts

- No, for no contacts

17. Was the forensic marking product(s) activated during any of these contacts?
    - Yes, the spray
    - Yes, the gel / grease
    - No
    - Not sure
18. It's really important that we hear from people who have had the device activated – if you have used the spray (if you have this) or if you suspect the gel (if you have this) has transferred to the person who has been told to stay away from you.

    Please add your telephone number or email address if you are happy for us to contact you about this.
19. Is there anything else you'd like to tell us about having the forensic marking product(s)?

    We hope that this research will improve police responses. We are also hoping to speak to some people in more depth by phone. Those who participate, even if you have already spoken to a researcher, will receive a £15 voucher to recognise their additional contribution to the research. For further details, please click here for the **participant information sheet**.
20. If you are interested in talking to a researcher in more depth about your experiences with the forensic marking product(s), please leave your mobile number and / or email address for us to contact you with more details.

    Before we outline the steps to receive your voucher, we would like to ask some questions about yourself, so we know we have spoken to a range of people.
21. What is your age group?
    - Under 26
    - 26-40
    - 41-55
    - Over 55
    - Prefer not to say

22. What is your ethnic group? Choose one section, then choose one option that best describes your ethnic group or background

- White
  - White – English/Welsh/Scottish/ Northern Irish/British
  - White – Irish
  - White – Gypsy or Irish Traveller
  - White – Roma
  - Any other White background, please describe
- Mixed/Multiple ethnic groups
  - Mixed/Multiple ethnic groups – White and Black Caribbean
  - Mixed/Multiple ethnic groups – White and Black African
  - Mixed/Multiple ethnic groups – White and Asian
  - Any other Mixed/Multiple ethnic background, please describe
- Asian/Asian British
  - Asian/Asian British – English/Welsh/Scottish/Northern Irish/British
  - Asian/Asian British – Indian
  - Asian/Asian British – Pakistani
  - Asian/Asian British – Bangladeshi
  - Asian/Asian British – Chinese
  - Any other Asian background, please describe
- Black/African/Caribbean/Black British
  - Black/African/Caribbean/Black British – English/Welsh/Scottish/Northern Irish/British
  - Black/African/Caribbean/Black British – African
  - Black/African/Caribbean/Black British – Caribbean
  - Any other Black/African/Caribbean background, please describe
- Other ethnic group
  - Other – Arab
  - Any other ethnic group, please describe

- Prefer not to say
  - Prefer not to say
- Please describe:

23. Disability is a physical or mental impairment, which has a substantial and long-term adverse effect on a person's ability to carry out normal day-to-day activities. This includes progressive and long-term conditions from the point of diagnosis, such as HIV, multiple sclerosis, cancer, mental illness or mental health problems, learning disabilities, dyslexia, diabetes, and epilepsy. This also includes 'disabled' as per the definition set out in the Equality Act 2010 and neurodiversity.

    Do you consider yourself to have a disability according to the definition above?

    - Yes
    - No
    - Prefer not to say

24. What is your sex?
    - Female
    - Male
    - Intersex
    - Prefer not to say

25. Is the gender you identify with, the same as your sex registered at birth?
    - Yes
    - No
    - Prefer not to say

26. Which of the following options best describes your sexual orientation?
    - Straight/Heterosexual
    - Gay or Lesbian
    - Bisexual
    - Prefer not to say
    - Other (please specify):

27. What is your religion?

- No religion
- Christian (including Church of England, Catholic, Protestant and all other Christian denominations)
- Buddhist
- Hindu
- Jewish
- Muslim
- Pagan
- Sikh
- Prefer not to say
- Other (please specify):

Thank you again for taking part in this research. You will receive a text within the next 14 days with your voucher details. To contact us about the evaluation or if you do not receive your voucher please email **forensic.marking@college.police.uk**

The research will be published in 2025 on the **College of Policing website**.

Appendix 4: Participant information sheet – forensic marking interviews (victims)

The College of Policing is the professional body for the police in England and Wales. The purpose of the College is to provide those working in policing with the skills and knowledge necessary to prevent crime, protect the public and secure public trust. To enable us to fulfil our purpose, we rely on volunteers to participate in research.

For this project, we are working in partnership with researchers from the Durham University Centre for Research into Violence and Abuse (CRiVA).

We would like to invite you to take part in one of our research projects. This information sheet tells you why the research is being done, what it will involve for you, what information we will ask from you, and what we will do with that information.

We will, in the course of this project, be collecting personal information. Under the General Data Protection Regulation (2016), we are required to provide a justification (what is called a “legal basis”) in order to collect such information. The legal basis for this project is “task carried out in the public interest”.

What is the purpose of the research?

We want to evaluate a forensic marking intervention to see how effective it is. People who have experienced domestic abuse are provided with products by the police (such as a handheld spray or gel), which can forensically mark a perpetrator to provide evidence that they have been present at a specific location. Perpetrators receive a warning that the victim has received a forensic marking kit, to deter them from offending again. The research is funded by the Cabinet Office.

Why have I been invited to take part?

You have received this invitation as you opted to find out more about taking part in an interview in the survey about your forensic marking products that you were given by the police.

What will happen if I agree to take part?

It is up to you to decide whether or not you want to take part in an interview. If you do decide to take part, you will be asked to sign a consent form saying you agree to participate.

We want to talk to people in each area who have received the forensic marking products to find out what they think about the products and their experience of the police. We would like you to take part in two interviews will be held on Microsoft Teams or by telephone, the first one approximately four weeks after receiving the handheld spray and/or gel and the second one four or five months later. These interviews will each last between 15 and 30 minutes. The interviews will be recorded to assist with analysis.

What are the benefits for me in taking part?

Your involvement will help us to understand if using forensic marking products is an effective way of providing support for people who have experienced domestic abuse and preventing further harm.

After the interview, you will receive a £15 voucher to recognise your contribution to the research.

Are there any risks for me if I take part?

The research is unlikely to pose any risk or disadvantage to you. The questions relate to what you think about the products you received, whether you have had any unwanted contact with the perpetrator and your experience of dealing with the police. If you did find that reflecting back causes you any distress, contact details for support services are listed at the end of the information sheet.

If you change your mind about participating in the research, you can withdraw your data up to seven days after you have completed the survey by sending your request to **forensic.marking@college.police.uk** and citing the last four digits of your mobile phone number.

What will you do with my information?

Any information you provide will be treated as confidential and research participants will be anonymised in any reports of findings from the research, so you will not be identified. The exception to this is where you tell us something that indicates that you or someone else is at risk of harm. In this instance, we may need to share this information with a relevant authority. However, we would speak to you about this first.

At the completion of the project, we will retain anonymised data from the research, which will be stored securely on College and Durham University systems for 10 years, in line with College policies.

We will use the data collected in this research to inform policing policy and practice, through the preparation of research reports, presentations and academic outputs, such as books or articles. We will place publicly accessible reports and academic outputs on the College and Durham University websites.

What happens next?

A researcher will contact you using the contact details you provided (either by text or email) to arrange a convenient time for you to take part in the interview.

If you have any questions, please contact Rosie Erol at **forensic.marking@college.police.uk**.

Support organisations

Victim Support: **www.victimsupport.org.uk** or 0808 1689 111

Women's Aid: **www.womensaid.org.uk**

National Domestic Abuse Helpline: 0808 2000 247

Men's Advice Line: 0808 801 0327

Appendix 5: Interview schedule – victims (time 1)

Consent process

- Have you had the opportunity to read the participant information sheet?
  - (If not, I can go over it with you now)
- Are you happy for me to take some notes during our phone call?
- Are you happy to participate in the interview?

A. Background

1. Could you tell me the brief circumstances that led to you been given the forensic marking products (the spray or the gel)?
2. Did they give you any warning stickers or posters?
3. Were you in touch with anyone else, any other support or organisations?
4. [How long ago were you given the spray?]

B. Receiving the intervention

5. Tell us about who gave you the product, and how they explained it to you.
6. Do you know what he's been told about it?
7. Would you feel confident if you had to use it?
8. Do you feel any different having it, even if you don't need to use it?
9. Could they have done anything better?
10. What do you think the main benefits might be to you having it?
11. What do you think the impact will be on your ex-partner?
12. Have they tried to contact you since you got the product?
    - In what ways?
13. Are there any potential disadvantages that you can think of?

End of interview.

Appendix 6: Interview schedule – victims (time 2)

Consent process

- Have you had the opportunity to read the participant information sheet?
  - (If not, I can go over it with you now)
- Are you happy for me to take some notes during our phone call?
- Are you happy to participate in the interview?

A. Background – what's changed?

1. Do you still have the product?
2. Has your life changed since the time we last spoke and you had just been given the forensic marking products (the spray or the gel)?
3. Did the product have any influence on these changes?

B. Living with the product

4. Did your feelings change about the product at all as time went on?
   - Did you continue to take it with you?
5. Have they tried to contact you since you got the product?
   - In what ways?
6. Did you have to use it or come close to using it?

C. Trust and confidence in policing

7. Did having the product make you feel any differently towards the police and how they respond to people who have experienced domestic abuse?

D. Finishing up

8. Overall, would you recommend this product to someone else who was experiencing domestic abuse?
9. What are the other things that would need to be in place as well as the product?
10. Is there anything else you would like to tell us that we haven't asked about?

End of interview.

Appendix 7: Participant information sheet – Police interviews

The College of Policing is the professional body for the police in England and Wales. The purpose of the College is to provide those working in policing with the skills and knowledge necessary to prevent crime, protect the public and secure public trust. To enable us to fulfil our purpose, we rely on volunteers to participate in research.

We would like to invite you to take part in one of our research projects. This information sheet tells you why the research is being done, what it will involve for you, what information we will ask from you, and what we will do with that information.

We will, in the course of this project, be collecting personal information. Under the General Data Protection Regulation 2016, we are required to provide a justification (what is called a “legal basis”) in order to collect such information. The legal basis for this project is “task carried out in the public interest”.

What is the purpose of the research?

We want to evaluate a forensic marking intervention to see how effective it is. People who have experienced domestic abuse are provided with products by the police (such as a handheld spray or gel), which can forensically mark a perpetrator to provide evidence that they have been present at a specific location. Perpetrators receive a warning that the victim has received a forensic marking kit, to deter them from offending again. The research is funded by the Cabinet Office.

Who is undertaking the research?

The evaluation is being conducted by a team of researchers who work for the College of Policing with support from researchers at Durham University.

Why have I been invited to take part?

You have received this invitation as you work for a police force and provide support for victims of domestic abuse. We are interested to understand what professionals who work with victims think about this intervention and how it is being implemented.

What will happen if I agree to take part?

It is up to you to decide whether or not you want to take part in an interview. If you do decide to take part, you will be asked to provide consent saying you agree to participate.

We want to talk to police and stakeholders who are involved in implementing the intervention at a local level, and/or developing policy and practice to support victims of domestic abuse. If you agree to take part, a researcher from the College will contact you to arrange an interview to be held in person or on Microsoft Teams. These interviews will each last up to 45 minutes. The interviews will be recorded and transcribed to assist with analysis.

What are the benefits for me in taking part?

Your involvement will help us to understand if using forensic marking products is an effective way of providing support for people who have experienced domestic abuse and preventing further harm.

Are there any risks for me if I take part?

The research is unlikely to pose any risk or disadvantage to you, other than the time taken to undertake the interview. If you did find that the interview causes you any distress, we would advise you to take up support offered by your organisation.

What will you do with my information?

Any information you provide will be treated as confidential and only members of the research team working on the project will be able to access the data. Research participants will be anonymised in any reports of findings from the research, so you will not be identified. The exception to this is where you tell us something that indicates that you or someone else is at risk of harm. In this instance, we may need to share this information with a relevant authority. However, we would speak to you about this first.

Personal data and interview transcripts will be securely stored on College and Durham University IT systems. Personal data will be deleted at the end of the project, which is expected to be March 2025. Any quotes used in the research report will be labelled using a pseudonym and will not include any detail that may identify

you. At the completion of the project, we will retain anonymised data from the research, which will be stored securely on College systems for 10 years, in line with College policies.

If you have any questions, please contact Rosie Erol at **rosie.erol@college.police.uk**

Appendix 8: Interview schedule – police interviews

Introduction

- Check received and read the participant information sheet.
- Request written consent to be interviewed and recorded if not already received [and confirm with verbal consent if written consent already provided].
- Explain the Cabinet Office Evaluation Accelerator Fund projects and the College role in these.
- Any questions before we begin?

Background

- Can you explain a bit more about the organisation:
  - What is your role within the organisation?
  - How does this fit with other departments, teams or districts involved in forensic marking?

Involvement and role in the intervention

- Please describe your involvement in implementing the intervention.
  - How and why did you get involved?
  - How long have you been involved?
- What were you overall thoughts about the intervention when you first heard about it being used in this context?
  - Has this changed at all over the time you've been involved?

Decision making about distributing the products

- How are decisions made about who received the forensic marking products?
  - Sign-off from senior officers
  - Recording decision making
- Who should be making the decision about interventions offered to domestic abuse victims?
  - Police, independent domestic violence advisers (IDVAs), multi-agency risk assessment conference (MARAC)?

- What factors do you think the police need to take into consideration when deciding who should receive these products?
- Does everyone who is suitable for receiving the products get offered one?
- Who do you think these products would work best for?
    - Are there any groups or characteristics that you think should not be receiving this type of intervention?
    - Any domestic-abuse-related crime types that are more or less suitable for this intervention?
    - Decisions about which products are provided?
    - What other products, interventions or support are offered to victims?

Implementing the intervention in practice

- Has the intervention been implemented as it was originally intended?
    - If not, what changes and why?
- How do you engage with victims who are selected to receive the intervention – what's the process for contacting them?
    - Does this work well?
    - Would you change anything about the process?
    - What proportion of victims take up the intervention?
    - What reasons do they give for not wanting to engage?
- How do you engage with suspects connected to the victims who are selected to receive the intervention – what's the process for contacting them?
    - What happens during the interaction?
    - Does this work well?
    - Would you change anything about the process?
- What has worked well with this intervention, in terms of putting it into practice?
- Have there been any challenges in setting up and implementing the intervention?
    - Have these challenges been overcome? How?
    - What was the overall impact of the challenges in being able to implement the intervention as intended?

- Ongoing monitoring and support to victims:
  - Who is involved?
  - How is this done?
  - Any challenges with data collection and monitoring?

Impact of the intervention

- What do you think the impact of the intervention is on victims?
  - Explore: feelings of safety, trust and confidence in the police, evidence of unwanted contact, deterrent effect.
- What do you think the effect will be on perpetrators of domestic abuse?
  - Explore: deterrent effect, changing methods of contacting the victim, reduction (or escalation) of harm.
- Are there any examples of cases where you think this intervention has been particularly effective?
  - Describe the situation, what made them successful or effective?

Overall comments

- Would you consider that the forensic marking intervention has been successful?
  - Please explain.
- What would you do differently if you were starting again, or advising another force to implement this intervention?
- Based on your experience, do you think there have been any unintended consequences from this intervention?
- Do you think this is an intervention that you/your organisation would [continue to] support?
- Anything further to add about the intervention (or the evaluation more generally)?

Additional questions for activation cases

Details of the case

- Thinking about the recent case where there was an activation of SmartWater, can you tell me a bit more about the background to the case?
    - Victim circumstances – why was the victim given the forensic marking product? Which products? Relationship with suspect.
    - Suspect – had the suspect been issued with a warning letter?
    - Previous incidents or offending.
    - Protection orders or bail conditions in place.
- Can you talk me through the details of the case and about your involvement in this?
    - Why and when were the police called?
    - What offence was reported?
    - When did you get involved?
    - What was your role?
    - What response did the police provide?
    - What happened at the scene [for example, evidence gathering, arrest, custody, interview, charging, support for the victim]?
    - What processes were followed that related specifically to the forensic marking activation?
    - Did the victim say anything about the forensic marking products?
    - What was the suspects reaction to the forensic marking?
    - What happens now for the victim?
    - What was the outcome?
    - How much time did you spend on this in relation to the forensic marking in particular [for example, explaining about the product, issuing suspect warning, processing in custody, taking samples for matching, and so on]?

Reflecting on what happened

- To what extent did the forensic marking product influence the outcome of the case?
- Did the processes around the forensic marking intervention work as expected?
- Is there anything that could have been done differently?
- Would you consider that the forensic marking intervention has been successful?
    - Please explain.
    - What are the key factors in helping it to work?
- Do you have any concern about the forensic marking, or are there any barriers that make it difficult to implement?
- Anything further to add about the case or the intervention more generally?

Appendix 9: Participant information sheet – Non-police stakeholder Interviews

The College of Policing is the professional body for the police in England and Wales. The purpose of the College is to provide those working in policing with the skills and knowledge necessary to prevent crime, protect the public and secure public trust. To enable us to fulfil our purpose, we rely on volunteers to participate in research.

We would like to invite you to take part in one of our research projects. This information sheet tells you why the research is being done, what it will involve for you, what information we will ask from you, and what we will do with that information.

We will, in the course of this project, be collecting personal information. Under the General Data Protection Regulation 2016, we are required to provide a justification (what is called a “legal basis”) in order to collect such information. The legal basis for this project is “task carried out in the public interest”.

What is the purpose of the research?

We want to evaluate a forensic marking intervention to see how effective it is. People who have experienced domestic abuse are provided with products by the police (such as a handheld spray or gel), which can forensically mark a perpetrator to provide evidence that they have been present at a specific location. Perpetrators receive a warning that the victim has received a forensic marking kit, to deter them from offending again. The research is funded by the Cabinet Office.

Who is undertaking the research?

The evaluation is being conducted by a team of researchers who work for the College of Policing with support from researchers at Durham University.

Why have I been invited to take part?

You have received this invitation as you work for an organisation that provides support for victims of domestic abuse. We are interested to understand what professionals who work with victims think about this intervention and how it is being implemented.

What will happen if I agree to take part?

It is up to you to decide whether or not you want to take part in an interview. If you do decide to take part, you will be asked to provide consent saying you agree to participate.

We want to talk to police and stakeholders who are involved in implementing the intervention at a local level, and/or developing policy and practice to support victims of domestic abuse. If you agree to take part, a researcher from the College will contact you to arrange an interview to be held in person or on Microsoft Teams. These interviews will each last up to 45 minutes. The interviews will be recorded and transcribed to assist with analysis.

What are the benefits for me in taking part?

Your involvement will help us to understand if using forensic marking products is an effective way of providing support for people who have experienced domestic abuse and preventing further harm.

Are there any risks for me if I take part?

The research is unlikely to pose any risk or disadvantage to you, other than the time taken to undertake the interview. If you did find that the interview causes you any distress, we would advise you to take up support offered by your organisation.

What will you do with my information?

Any information you provide will be treated as confidential and only members of the research team working on the project will be able to access the data. Research participants will be anonymised in any reports of findings from the research, so you will not be identified. The exception to this is where you tell us something that indicates that you or someone else is at risk of harm. In this instance, we may need to share this information with a relevant authority. However, we would speak to you about this first.

Personal data and interview transcripts will be securely stored on College and Durham University IT systems. Personal data will be deleted at the end of the project, which is expected to be March 2025. Any quotes used in the research report will be labelled using a pseudonym and will not include any detail that may identify

you. At the completion of the project, we will retain anonymised data from the research, which will be stored securely on College systems for 10 years, in line with College policies.

If you have any questions, please contact Rosie Erol at **rosie.erol@college.police.uk**

Appendix 10: Interview schedule – non-police stakeholders

Introduction

- Check received and read the participant information sheet.
- Request written consent to be interviewed and recorded if not already received [and confirm with verbal consent if written consent already provided].
- Explain the Cabinet Office Evaluation Accelerator Fund projects and the College role in these.
- Any questions before we begin?
- [For external stakeholders] Explain what the intervention is – show video (or send link to this beforehand).

Background

- Can you explain a bit more about the organisation you work for and how you support victim-survivors or perpetrators of domestic abuse?
    - What is your role within the organisation?

Understanding of forensic marking

- Before contacting you, had you heard of forensic marking as an intervention to support domestic abuse victims?
    - Please explain.
- What are your initial thoughts about this as an intervention?
- Are you aware of forensic marking as an intervention for other crime types (not related to domestic abuse)?
- Have you had any direct experience of this intervention?

Benefits and concerns about the products

There are a number of products that are available as part of the forensic marking intervention. These are:

- the home protection pack (to mark products)
- warning stickers and posters

- handheld spray
- gel or grease
- plug-in spray or alarm system for inside the home.

In each case, the perpetrator will be warned by the police that the victim has received forensic marking products (though the actual product is not specified).

- Thinking about the actual products, what do you think are the benefits of each of these?
  - Home protection kits and warning stickers
  - Gel or grease
  - Handheld spray
  - Plug-in spray or alarm system
- Do you see any potential disadvantages of each of the products, or have any concerns about how they might be used or misused?
  - Home protection kits and warning stickers
  - Gel or grease
  - Handheld spray
  - Plug-in spray or alarm system

Expected impact of the intervention

- What do you think the impact of the intervention be on victims?
  - Explore: feelings of safety, trust and confidence in the police, evidence of unwanted contact, deterrent effect.
- What do you think the effect will be on perpetrators of domestic abuse?
  - Explore: deterrent effect, changing methods of contacting the victim, reduction (or escalation) of harm.
- What other support or interventions need to be in place for this to be effective?

Overall comments

- Do you think this is an intervention that you or your organisation would [continue to] support?
- Anything further to add about the intervention (or the evaluation more generally)?

Appendix 11: Characteristics of intervention group cases by force

The characteristics of people receiving the forensic marking products were largely similar across all four police forces. Table A1 shows the characteristics for the whole intervention group across all four forces that delivered the intervention. This includes the victims who were not included in the final impact analysis, following the matching process with the comparison group.

Most victims were female and most perpetrators were male. Victims were slightly younger than perpetrators on average, with median ages of 33 years and 35 years respectively. Information about ethnicity was inconsistently recorded, and was missing for all of the perpetrators from one force.

In terms of the products provided, most people received the spray canister. Home protection kits were predominantly given out in West Yorkshire. In total, 7% of victims had the grease applied to their properties, with most of these also receiving a spray. In 60% of cases, the perpetrator received a warning that the victim had been provided with forensic marking products, although there was no record of this in 11% of cases. Just under 60% of cases were assessed as being high-risk and one-third were assessed as being medium-risk. Slightly less than half of the people receiving forensic marking products had been the victim of stalking.

Table A1: Characteristics of forensic marking cases.

Characteristic	**Total**	**%**
Number of victims - IPV	288	100
Victim sex		
Male	5	2%
Female	283	98%
Victim ethnicity		
White	225	78%
Black	7	2%
Asian	11	4%

Characteristic	Total	%
Other	3	1%
Missing or not known	42	15%
Victim age at time of incident		
Median (min, max)	33	(17, 73)
Perpetrator sex		
Male	268	93%
Female	8	3%
Missing or not known	12	4%
Perpetrator ethnicity		
White	132	46%
Black	23	8%
Asian	21	7%
Missing or not known	112	39%
Perpetrator age at time of incident		
Median (min, max)	35	(17, 65)
Forensic marking products provided		
Home protection kit	74	26%
Grease	20	7%
Spray or canister	257	89%
Mobile unit	1	< 1%
Missing or not known	6	2%
Visible warning stickers provided		
Yes	177	61%
No	95	33%

Characteristic	Total	%
Missing or not known	16	6%
Warning delivered to perpetrator		
Yes	172	60%
No	85	30%
Missing or not known	31	11%
Risk assessment		
High	167	58%
Medium	101	35%
Standard	10	3%
Missing or not known	10	3%
Perpetrator arrested		
Yes	248	86%
No	33	11%
Missing or not known	7	2%
Perpetrator charged		
Yes	95	33%
No	181	63%
Missing or not known	12	4%
Protection order or bail conditions in place		
Yes	182	63%
No	82	28%
Missing or not known	24	8%
Perpetrator – number of offences over last 12 months		
0	113	39%

Characteristic	Total	%
1-2	89	31%
3-5	42	15%
6-10	35	12%
>10	5	2%
Missing or not known	4	1%
Incident type resulting in products being offered		
Stalking	125	43%
Breach	31	11%
Actual bodily harm	24	8%
Coercive or controlling behaviour	19	7%
Non-fatal strangulation	14	5%
Rape	13	5%
Criminal damage	9	3%
Common assault	8	3%
Grievous bodily harm with intent	7	2%
Harassment	6	2%
Threats to kill	6	2%
Other	13	6%
Missing or not known	12	4%

Fewer cases were used in the final model, as this only included data from three forces, and focused on female victims and cases assessed as being at high-risk or medium-risk.

The characteristics of the intervention group and comparison group used in the quantitative analysis are presented in **Appendix 13**.

Appendix 12: Technical annex

Propensity score matching

Comparison data was formed using incident records from the participating police forces, which were collected during a two-week period during which the intervention was not being used. Within each force, a logistic regression model was used to create a propensity score. The outcome of this model was whether or not the participant had been given some forensic marking equipment (the variable 'FM_equip'). The explanatory variables were as follows (with their variable name given as in the table in **Appendix 13**):

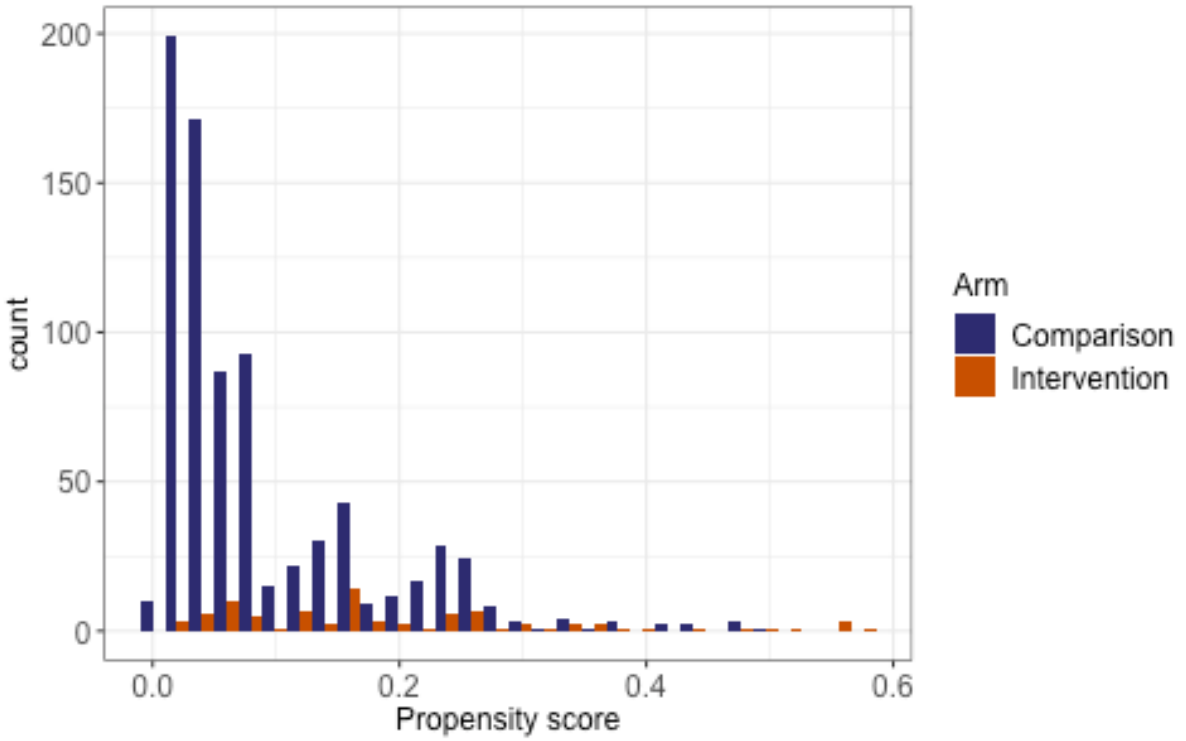
- Whether the reference incident was course of conduct ('CoC').
- Whether the reference incident was violent or sexual ('violent_SO').
- The risk assessment ('risk_ass').
- Whether the suspect had any course of conduct incidents in previous 12 months ('CoC_susprev').
- Whether the suspect had any violent or sexual incidents in previous 12 months ('violent_SO_susprev').
- The number of incidents relating to the suspect in the previous 12 months ('n_prev_sus').
- Whether a protection order was in place ('protection_order').

Risk assessment is carried out by the police to assess the level of threat of harm posed to the victim by the perpetrator. Medium-risk cases show a pattern of abuse or control. There is a potential for further harm, but this is unlikely to happen without a change in circumstances. For cases assessed as being high-risk, the perpetrator poses a serious threat of harm to the victim, which could happen at any time and would have a serious impact.

The propensity score matching was done in two stages, because protection order data could only be retrieved for a smaller number of cases. An initial logistic regression model was therefore fitted using all explanatory variables except 'protection_order'. This model was used to generate a subset for which the propensity scores had a similar range and distribution to the scores of the intervention group. Sometimes, this involved 'thinning' the comparison group over

certain ranges of propensity score. For example, for West Yorkshire, a random sample of 80 incidents was generated from those entries with a propensity score below 0.08. All records were kept for those with initial propensity scores greater than or equal to 0.08. This reduced group of potential comparison incidents was then sent back to the force, so that they could look up protection order information.

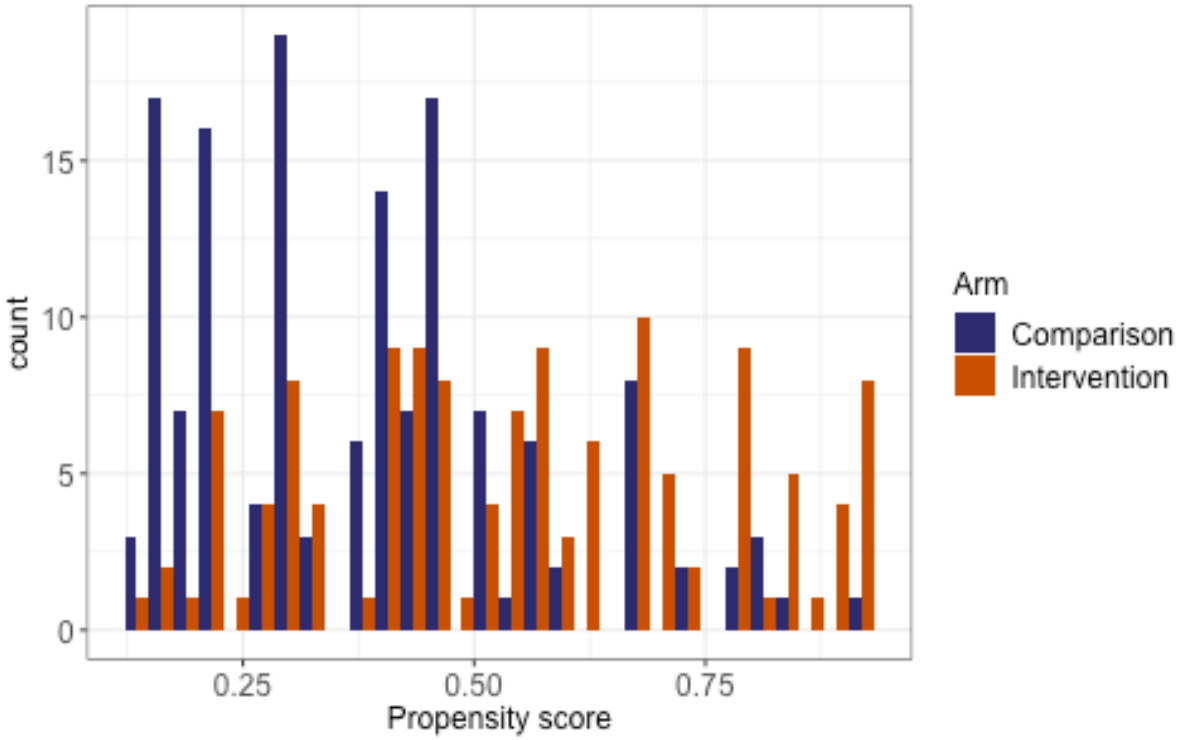
Figure A1: Initial fitted propensity scores for West Yorkshire data.



Once the protection order data was known, the logistic regression model was refitted to include this explanatory variable as well as the others.

For all forces, there were a substantial number of comparison cases with propensity scores lower than the lowest propensity score in the intervention group. These were discarded (see Figure A1 and Figure A2 for examples of scores for West Yorkshire).

Figure A2: Refitted propensity scores for West Yorkshire data.



In general, there tended to be a greater number of high propensity scores in the intervention group and a greater number of low propensity scores in the comparison group. To create balanced groups, random 'balance sampling' was used.

Balance sampling

To create balanced comparison and intervention groups, the following process was used (for each force separately):

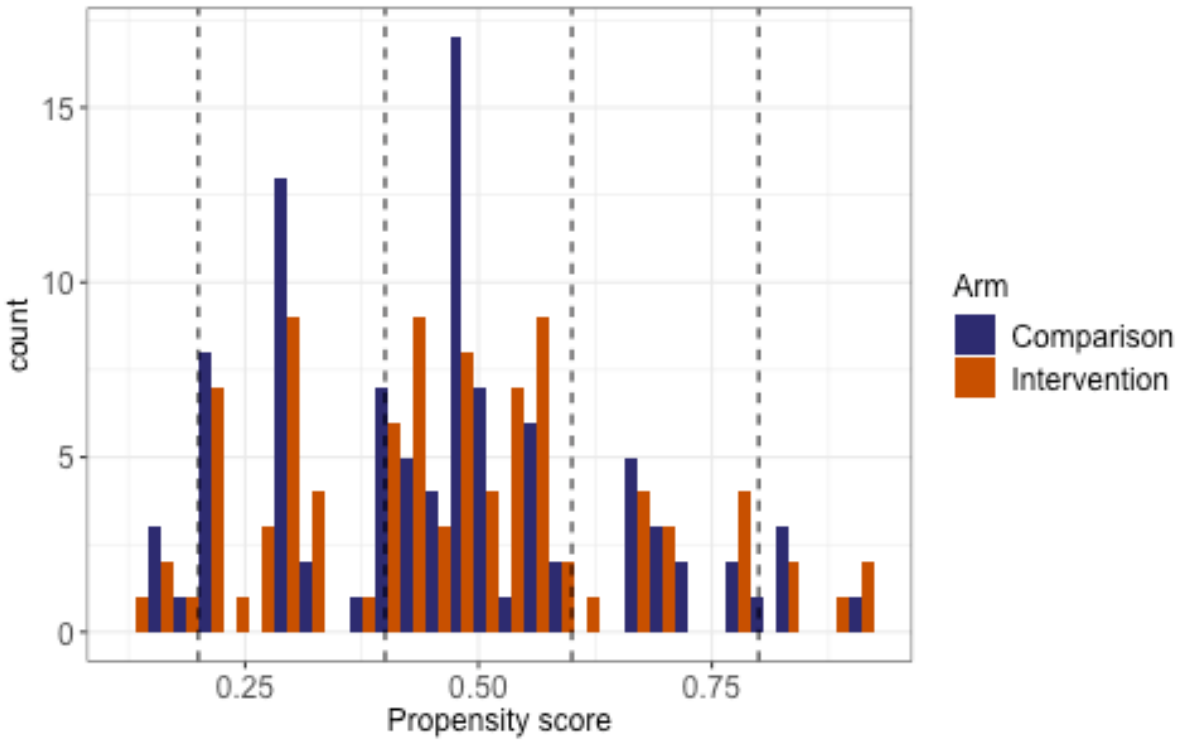
- Split the range [0,1] into n equal intervals.
- For each interval, count the number of intervention group rows and number of comparison group rows with propensity score in that range.
- Whichever group is larger or more represented in that bin, sample the number of the other group from those rows. This way, we have equal numbers of comparison and intervention rows with propensity score in that range.

For Durham and West Yorkshire, five bins were used (so splitting propensity score into intervals of width 0.2). For Hampshire, four bins were used (so bins of width 0.25). Figure A3 shows a balance sampled data set for West Yorkshire. It has 188 rows (94 each for comparison and intervention). The dashed lines show the divisions

between the propensity score bins. Between each pair of dashed lines, there are equal numbers of comparison and intervention cases.

Because covariates are adjusted for in the final model, it is not necessary to have exact balance. However, some level of balance is useful so as not to over-rely on parametric assumptions.

Figure A3: Balance sampling of propensity scores for West Yorkshire data.



Impact analysis

Modelling strategy

A Bayesian negative binomial random intercept model was used for the number of follow-up events, adjusted for the same explanatory variables as the propensity score model.

A random intercept was included to account for differences between police forces. A negative binomial distribution was used because the outcome (number of follow-up events), was over-dispersed count data.

The models were fitted to each of Durham and West Yorkshire separately, to ensure that the assumption of using the same coefficients was reasonable, which appeared

to be the case. There was not enough data from Hampshire to fit a model to that force on its own.

To get to the final model, the variables included to start with were:

- FM_equip
- CoC
- violent_SO
- risk assessment
- protection order
- number of previous incidents
- any CoC previous
- any violent_SO previous
- a random effect for police force

Initial fitted models were used to remove terms that led to a variance inflation factor greater than 4. The least significant terms were also removed. The model was refitted and the process was repeated until 0 was not in the central 50% for any terms. This is essentially a backwards elimination stepwise selection process, but being quite generous with which terms are kept.

The terms that were kept were:

- protection_order
- nprev_sus
- CoC_susprev
- violent_SO_susprev
- the random effect for police_force

Sensitivity analysis to balance sampling

The balance sampling process is random, and so could be repeated many times to produce different final data sets. A seed in the function was used to ensure replicability.

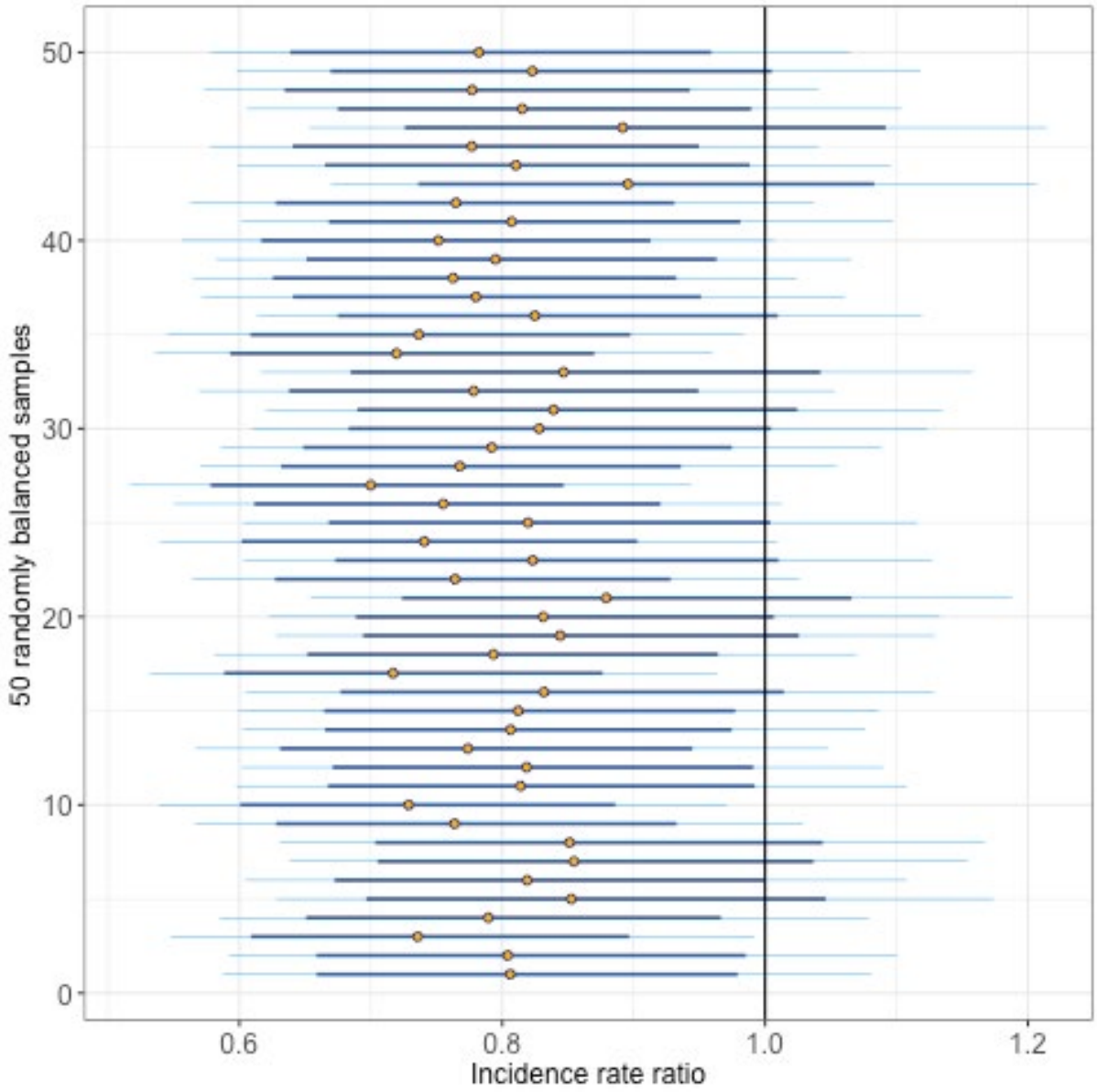
Once the model was finalised (of number of follow-up events in relation to the other factors) the following process was used for i from 1 to 50:

- Balance sample with seed set to i.
- Fit the model to the resulting data set.
- Record the 2.5%, 10%, 50%, 90% and 97.5% quantiles of the treatment effect, the coefficient of FM_equip.

Results

Figure A4 shows the results of the experiment described in the previous section.

Figure A4: Median (orange dots), central 80% (dark blue lines) and central 95% (light blue lines) for posterior samples of incidence rate ratio using the model described.



From this, we see that the model is sensitive to the balance sampling. However, there is compelling evidence that the incidence rate ratio is less than 1.

Code

Example of logistic model fit, where 'full_data' contains all intervention and potential comparison data for a particular force.

```
psm_1 = glm(
 FM_equip ~ CoC + violent_SO + risk_ass + CoC_susprev +
violent_SO_susprev + nprev_sus,
 data = full_data,
 family = binomial(link="logit"))
```

- Add initial propensity scores into full_data

```
fitted_po = predict(
 object = psm_1,
 newdata = full_data,
 type = "response"
 )
 full_data$propensity_score1 = fitted_po
```

- Remove comparison group cases with propensity score below minimum for intervention group.

```
min_po_int = min(full_data$propensity_score1[full_data$FM_equip])
 full_data_pomin =
full_data[(full_data$propensity_score1>=min_po_int),]
```

- More comparison cases may be removed – for example, if thinning is needed in some regions to reduce the number of protection order look-ups.
- 'full_data_po' is the data set with all intervention cases and those comparison cases that remain after the initial propensity score modelling process.
- Re-fit propensity score model once protection order is available.

```
psm_2 = glm(
 FM_equip ~ CoC + violent_SO + risk_ass + CoC_susprev +
violent_SO_susprev +
```

```
 nprev_sus + protection_order,
 data = full_data_po,
 family = binomial(link="logit"))
# Add final propensity scores into full_data_po
final_po = predict(
 object = psm_2,
 newdata = full_data_po,
 type = "response"
 )
 full_data_po$propensity_score_final = final_po
```

- The function 'fm_balance' creates a subset of full_data_po that is balanced in terms of FM_equip within each propensity score bin.

```
library(tidyverse)
fm_balance = function(
 data, # full dataset, 'full_data_po' in this example code
 seed = 1, # random seed, for reproducibility
 balance_col = "FM_equip", # column to balance (intervention)
 balance_score = "ps_final", # score to balance by (propensity
score)
 id_col = "Incident_ID", # ID variable
 n_bins = 5 # number of bins to use for balancing
 ){
 set.seed(seed)
 bins_seq = seq(from=0, to=1, length= n_bins+1)
 n_data = nrow(data)
 ps_vec = data[[balance_score]]
 fm_vec = data[[balance_col]]
 id_vec = data[[id_col]]
 data_samp = data[numeric(0), ]
 for (i in 1:n_bins){
 ps_range = bins_seq[i:(i+1)]
 data_bin_i = data[(between(ps_vec, ps_range[1], ps_range[2])),]
 data_bin_fm = data_bin_i[data_bin_i[[balance_col]] == TRUE, ]
```

```
data_bin_comp = data_bin_i[data_bin_i[[balance_col]] == FALSE, ]
fm_count_i = nrow(data_bin_fm)
comp_count_i = nrow(data_bin_comp)
if(fm_count_i == comp_count_i){
data_samp = rbind(data_samp, data_bin_i)
} else if (fm_count_i > comp_count_i){
## Sample comp_count_i rows from the fm data
## add to data frame
fm_samp_i = data_bin_fm[
sample(
x=1:fm_count_i,
size = comp_count_i
),
]
data_samp = rbind(data_samp, data_bin_comp, fm_samp_i)
} else if (comp_count_i > fm_count_i){
## Sample fm_found_i rows from the comp data
## add to data frame
comp_samp_i = data_bin_comp[
sample(
x=1:comp_count_i,
size = fm_count_i
),
]
data_samp = rbind(data_samp, data_bin_fm, comp_samp_i)
}
return(data_samp)
}
```

- For the impact analysis, the negative binomial model is fit to balance sampled data sets, using the 'stan_glmer' function and its default weakly informative priors. The balance sampling is done separately for each force.

```
wdh_rep_bal = function(
 seed,
```

```
model_formula = "n_fo_dyad ~ FM_equip + protection_order +
nprev_sus +
CoC_susprev + violent_SO_susprev + (1|police_force)",
irr=T # Return incident rate ratios
){
dur_bal = fm_balance(data=wdh_df[wdh_df$police_force == "Durham",
], n_bins = 5, seed=seed)
wy_bal = fm_balance(data=wdh_df[wdh_df$police_force == "WYP", ],
n_bins = 5, seed=seed)
h_bal = fm_balance(data=wdh_df[wdh_df$police_force == "Hampshire",
], n_bins = 4, seed=seed)
wdh_bal = rbind(dur_bal, wy_bal, h_bal)
 glm_wdh = stan_glmer(
eval(model_formula),
 data = wdh_bal,
family = neg_binomial_2,
QR=T,
adapt_delta = 0.99)

 perc2_5 = glm_wdh$stan_summary[2,"2.5%" ]
perc10 = glm_wdh$stan_summary[2,"10%" ]
median = glm_wdh$stan_summary[2,"50%" ]
perc90 = glm_wdh$stan_summary[2,"90%" ]
perc97_5 = glm_wdh$stan_summary[2,"97.5%" ]

 if(irr){
return(
data.frame(
perc2_5 = exp(perc2_5),
perc10 = exp(perc10),
median = exp(median),
perc90 = exp(perc90),
perc97_5 = exp(perc97_5)
```

```
)
)
 } else {
return(
data.frame(
perc2_5 = perc2_5,
perc10 = perc10,
median = median,
perc90 = perc90,
perc97_5 = perc97_5
)
)
 }
}
```

- This process is repeated 50 times, with seeds from 1 to 50, to generate the results shown in Figure A3.

```
df_wdh_balrep = wdh_rep_bal(seed=1)
for (i in 2:50){
row_i = wdh_rep_bal(seed=i)
df_wdh_balrep = rbind(df_wdh_balrep, row_i)
}
```

Appendix 13: Characteristics of the intervention and comparison groups

Variable	**Description**	**Intervention** (n = 232)	**Comparison** (n = 276)
Characteristics		**n (%)** unless noted	**n (%)** unless noted
victim_sex	Female	232 (100%)	276 (100%)
suspect_sex	Female	3 (1.3%)	4 (1.4%)
	Male	223 (96%)	264 (96%)
	Other or not specified	6 (2.6%)	8 (2.9%)
suspect_age	Age of suspect – median (Q1, Q3)	35 (29, 41)	36 (30, 42)
police_force	Durham	87 (38%)	119 (43%)
	Hampshire	22 (9.5%)	17 (6.2%)
	West Yorkshire	123 (53%)	140 (51%)

Variable	Description	Intervention (n = 232)	Comparison (n = 276)
Reference crime			
CoC	Course of conduct	130 (56%)	134 (49%)
violent_SO	Violence or sexual offence	67 (29%)	104 (38%)
breach	Breach	24 (10%)	20 (7.2%)
theft_CD	Theft or criminal damage	8 (3.4%)	18 (6.5%)
non-crime	Non-crime	1 (0.4%)	0 (0%)
NA	Not known	2 (0.9%)	0 (0%)
FM_equip	Victim given forensic marking products	232 (100%)	0 (0%)
warning_YN	Warning given to perpetrator	132 (57%)	0 (0%)
risk_ass	**Risk assessment**		
	High-risk	148 (64%)	135 (49%)
	Medium-risk	84 (36%)	141 (51%)
	Standard-risk	0 (0%)	0 (0%)
protection_order	Protection order or bail conditions in place following the reference incident	140 (60%)	134 (49%)

Variable	Description	Intervention (n = 232)	Comparison (n = 276)
Offences over the previous 12 months			
nprev_sus	Number of offences by the perpetrator over previous 12 months – median (Q1, Q3)	1.00 (0.00, 3.00)	1.00 (0.00, 4.00)
CoC_susprev	Number of perpetrators with previous course of conduct offences	55 (24%)	78 (28%)
violent_SO_susprev	Number of perpetrators with previous violence or sexual offences	100 (43%)	166 (60%)
Offences over six-month follow-up period			
n_fo_sus	Number of offences by the perpetrator for the six-month follow-up period – median (Q1, Q3)	0.50 (0.00, 2.00)	1.00 (0.00, 2.00)
n_fo_dyad	Number of offences by the suspect for the six-month follow-up period with the same perpetrator-victim dyad as the reference offence – median (Q1, Q3)	0.00 (0.00, 1.00)	0.00 (0.00, 2.00)

Crime categories

The crime recorded for the intervention and comparison groups covered a large range of offences. These were recoded into five categories to help the analysis process. These are summarised as follows:

Category	Variable	Examples of offences included
Course of conduct offence	CoC	▪ Harassment, alarm or distress ▪ Coercive and controlling behaviour ▪ Disclose or threats to disclose private sexual photographs or film with intent to cause distress ▪ False imprisonment ▪ Fear or provocation of violence ▪ Sending letters etc with intent to cause distress or anxiety ▪ Stalking ▪ Threats to kill
Violence and sexual offences	violent_SO	▪ Assault occasioning actual bodily harm ▪ Cause serious injury by careless or inconsiderate driving ▪ Common assault ▪ Grievous bodily harm ▪ Malicious wounding ▪ Non-fatal strangulation ▪ Rape
Breach of a protection order or bail conditions	breach	▪ Breach of non-molestation order ▪ Breach of a restraining order ▪ Breach of bail, court order or injunction

Theft and criminal damage	theft_CD	▪ Arson ▪ Burglary residential ▪ Criminal damage ▪ Robbery ▪ Theft of a motor vehicle
Non-crime incident	non-crime	▪ Domestic incident ▪ No linked crime ▪ Non-crime domestic

About the College

We're the professional body for the police service in England and Wales.

Working together with everyone in policing, we share the skills and knowledge officers and staff need to prevent crime and keep people safe.

We set the standards in policing to build and preserve public trust and we help those in policing develop the expertise needed to meet the demands of today and prepare for the challenges of the future.

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